



Novel Machine Learning Methods for Process Optimization Problems in Biological
Wastewater Treatment

by

Egberto F. Selerio Jr.

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1 Introduction

Water pollution presents a critical environmental and public health challenge worldwide, with particularly severe repercussions in developing nations (Lin et al., 2022; Madhav et al., 2020). In the Philippines, over 90% of wastewater is discharged untreated into natural water bodies annually (Tabios III, 2020). This leads to the degradation of aquatic ecosystems, which pose risks to human health (NEDA & DOH, 2010; Tabios III, 2020). Addressing this issue requires innovative and contextually appropriate wastewater treatment technologies. Biological wastewater treatment is the most widely used consortium of technologies worldwide to combat this issue (Duan et al., 2022; Hellal & Abou-Elela, 2021; Liang et al., 2021; Rapi et al., 2021; Winkler & Straka, 2019) and is the dominant approach applied for wastewater treatment in developing countries (Damalerio, Beltran, Orbecido, et al., 2022; Machdar et al., 1997; Ulrich et al., 2009). There are two objectives in biologically treating domestic wastewater. The first objective is to substantially reduce its organic matter content, which inhibits the growth of heterotrophic organisms (Marais & Ekama, 1976). This process prevents the potential spread of disease and mitigates the depletion of O_2 in water bodies, which is a common consequence of excessive heterotrophic activity. The secondary goal is to curtail the concentration of eutrophic substances, such as NH_4^+ , NO_2^- , and NO_3^- (Marais & Ekama, 1976). This curtailment restricts the growth of autotrophic organisms and safeguards the receiving water bodies from harmful algal blooms.

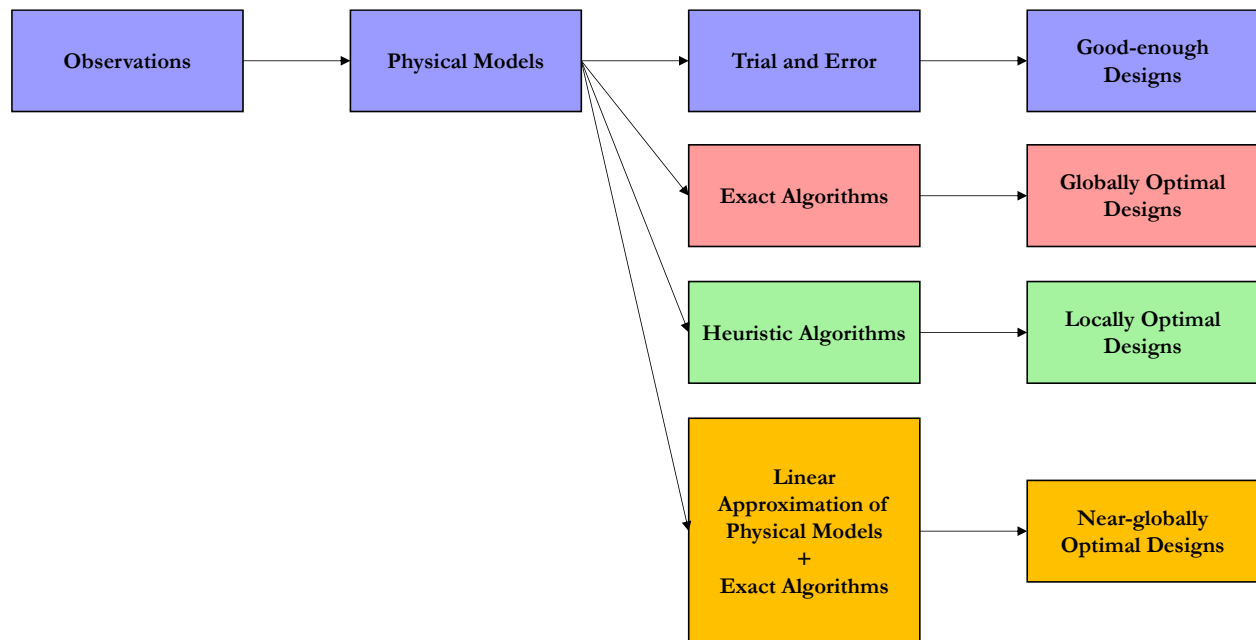


Figure 1-1 General wastewater treatment design flow with alternative computational methodologies and design outputs

Designing biological wastewater treatment systems involves a hierarchy of methodologies (Aboagye et al., 2021; Gu et al., 2023; Narayanan & Narayan, 2019; Padrón-Páez et al., 2020; Rivas et al., 2008; Shojaei & Shojaei, 2021), each with unique strengths and limitations. A general structure of the alternative computational design methodologies employed in the literature is summarized in **Figure 1-1**. At the most fundamental level, direct observations—such as monitoring microbial populations, nutrient concentrations, sludge quality, and oxygen demand—provide the baseline data necessary for any subsequent design work (Croll et al., 2023; Manav-Demir et al., 2024). These observations feed into biophysical models that simulate key biochemical processes (Henze et al., 2006). Widely recognized models, such as the activated sludge models (ASM) (Henze et al., 2006), make it possible to predict effluent quality (Calise et al., 2020; Szeląg et al., 2022), manage energy usage (Szeląg et al., 2022), and achieve process stability (Caro et al., 2024; Jafarinejad, 2020) in biological wastewater treatment systems. Initial design efforts often follow a trial-and-error approach, making minor adjustments like altering retention times, aeration rates, or pH levels, to achieve acceptable outcomes for the system (Sadri Moghaddam & Pirali, 2021). This approach is often assisted with simulation software like GPS-X, Biowin, AQUASIM, SUMO, and SIMBA (Bazemo et al., 2021; Bressani-Ribeiro et al., 2021; Caro et al., 2024; Jafarinejad, 2020; Sadri Moghaddam & Pirali, 2021).

However, as system complexity grows and optimization demands increase, exact algorithms can be employed to mathematically search for globally optimal configurations (Aboagye et al., 2021; Damalerio, Beltran, & Aviso, 2022; Padrón-Páez et al., 2020). In practice, computational challenges and non-linearities in biological reactions limit the feasibility of exact optimization algorithms, leading to the adoption of heuristic optimization algorithms—like genetic algorithms, simulated annealing, or generalized reduce gradient—that generate workable, locally optimal solutions (Gu et al., 2023; Jana et al., 2022; B. Li et al., 2022; Niu et al., 2022; Rivas et al., 2008; Shojaei & Shojaei, 2021). To bridge the gap between computational complexity and accuracy, linear approximations of non-linear models can be combined with exact algorithms, delivering near-globally optimal designs that approach ideal performance without excessive calculation times (Kim et al., 2009; Kusiak & Wei, 2012; M. Li et al., 2020; T. Yang et al., 2014). Ultimately, depending on design requirements and resource constraints, the selection of design methodology can yield outcomes ranging from basic “good-enough” solutions to near-globally optimal ones, creating a spectrum of approaches that wastewater engineers can strategically apply for the optimization of new or existing treatment infrastructure.

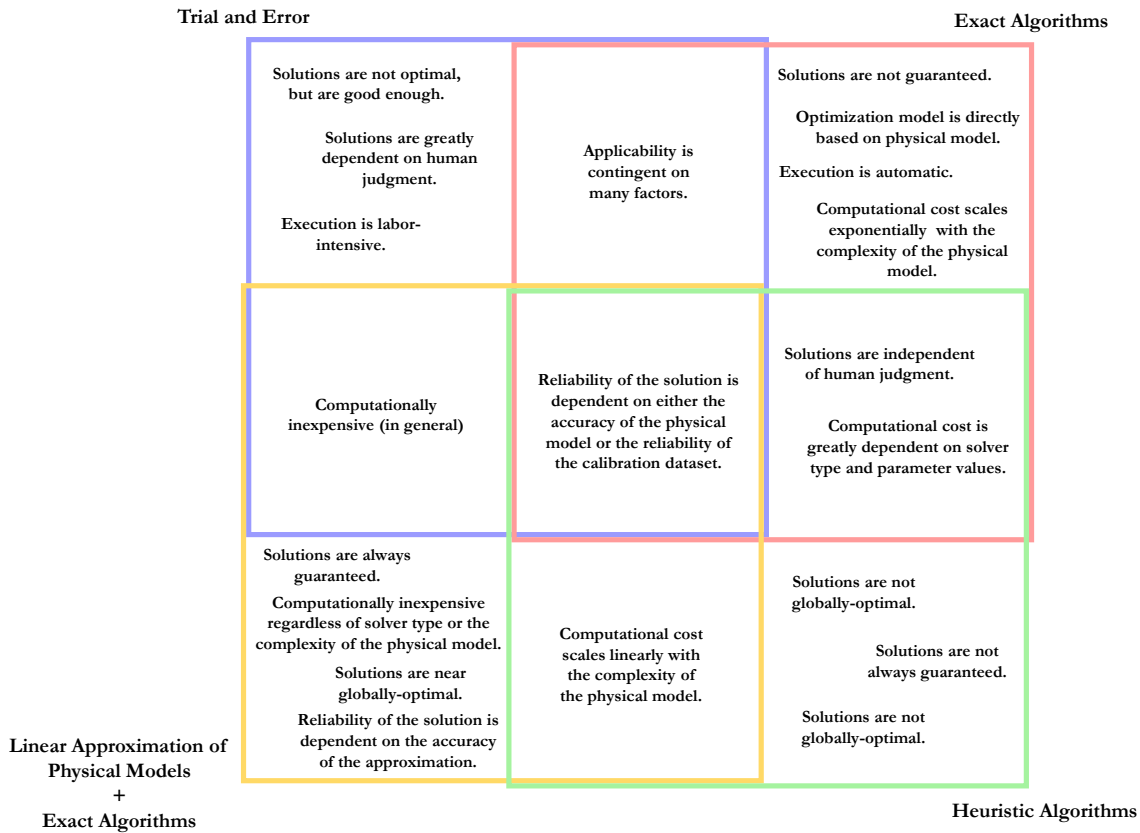


Figure 1-2 Similarities and differences between the computational design methodologies

A comparison of the alternative computational design approaches described above are presented in **Figure 1-2**. Trial-and-error methods excel at providing quick fixes and allowing operators to make incremental improvements based on empirical observations. It is the most commonly applied approach in practice with the use of various simulation software (Bazemo et al., 2021; Caro et al., 2024; Díez-Montero et al., 2019; Hellal & Abou-Elela, 2021; Jafarinejad, 2020; Jasim, 2020; Kasi et al., 2011; Lainé et al., 1999; Liang et al., 2021; Nazif et al., 2023; Sadri Moghaddam & Pirali, 2021; Simsek et al., 2012). Such methods are manageable, but they often require substantial human effort and are labor-intensive. On the opposite end, exact algorithms seek globally optimal configurations and ensure that no superior solution remains undiscovered in an optimization routine that seeks either optimal process design or control. Though computationally demanding, these algorithms provide high precision and are best suited for well-defined and smaller-scale problems, where linearity assumptions can be considered acceptable (Aboagye et al., 2021; Damalerio, Beltran, & Aviso, 2022; Padrón-Páez et al., 2020). However, this rarely happens when dealing directly with complex biochemical models of wastewater treatment processes, which are often nonlinear due to Monod functions that define reaction rates (Henze et al., 2006). Positioned between these extremes, heuristic algorithms offer a pragmatic compromise by applying educated “rules of thumb” to

navigate large, complex, and highly non-linear scenarios (Poh et al., 2016). While they may not guarantee a global optimum, they efficiently generate good solutions within reasonable timeframes, making them popular for intricate systems where exact approaches may be infeasible (Poh et al., 2016).

On the other hand, linear approximations present a way to simplify complex, non-linear biological processes into more manageable models (Kim et al., 2009; Kusiak & Wei, 2012; M. Li et al., 2020; T. Yang et al., 2014). These linear model approximations may directly be calibrated using data that would otherwise be used to calibrate the nonlinear biophysical models that are usually used in simulation software like Biowin, GPS-X, AQUASIM, SUMO, and SIMBA (Bressani-Ribeiro et al., 2021; Hellal & Abou-Elala, 2021; Kasi et al., 2011; Lainé et al., 1999; Liang et al., 2021). Unlike nonlinear biophysical models, these methods can be integrated with exact optimization algorithms to achieve near-globally optimal solutions without imposing excessive computational demands. This balanced approach effectively combines accuracy with practicality. However, a notable challenge is that linearization techniques commonly employed in the literature, such as Taylor series expansions (M. Li et al., 2020; T. Yang et al., 2014), significantly restrict the applicability of the models to the vicinity of the approximation point. Furthermore, while such approaches are frequently utilized in process controller design (M. Li et al., 2020; T. Yang et al., 2014), their application to optimal process design—such as wastewater network optimization and process set-point determination—remains relatively rare. Another linearization technique is through multivariable polynomial fitting (Ahn et al., 2014; Kim et al., 2009; Lim et al., 2011). However, the current application of this approach usually disregards the interdependence of compounds in biological wastewater treatment systems, which is a major flaw. Kusiak & Wei (2012) was the first to address this issue with a system of coupled nonlinear equations for the activated sludge process, but their training dataset included target variables which constitute a major data spillage. Their model is also nonlinear. At present, a competent linear model addressing the issues presented above is not yet developed.

On the other hand, common machine learning (ML) methods, such as XGBoost, LightGBM, random forests, support vector machines (SVM), and artificial neural networks (ANNs), while widely used in soft sensing and optimal control (Croll et al., 2023; Fang et al., 2011; Manav-Demir et al., 2024), are rarely utilized for approximating biophysical models of biological wastewater treatment systems like, for example, bio-towers (Tyagi et al., 2021). These ML methods yield nonlinear approximations, but they can still be solved with greater ease than the original biophysical models using heuristic optimization algorithms (Fang et al., 2011). For example, Fang et al. (2011) trained an SVM using simulation data then performed optimization using the trained SVM model. A similar data-driven application was performed by Asadi et al. (2017) but with random forests, artificial neural networks, multi-adaptive regression spline, and k-nearest neighbors algorithms. ML methods are suitable candidates for the approximation because they can

simultaneously perform predictions on multiple wastewater compounds. Multiple response optimization has been emphasized as an important research direction in the field of biological wastewater treatment (Nazlabadi et al., 2021).

On the other hand, ML method that generate linear models that can perform simultaneous predictions on multiple wastewater compounds is presently unexplored in the literature (Nazlabadi et al., 2021). An obvious candidate for such a method would be some new implementation of the canonical regression technique, but no attempt is currently made for applications such as approximating nonlinear biophysical models of biological wastewater treatment systems. This research direction is interesting because it can greatly improve on existing linear approximations like Taylor series expansion (M. Li et al., 2020; T. Yang et al., 2014) and multivariable polynomial fitting (Ahn et al., 2014; Kim et al., 2009; Lim et al., 2011); and would facilitate near-global optimization for either process design or control unlike the previously mentioned ML models which are only solvable using heuristic optimization algorithms that yield locally optimal solutions.

The special case of bio-towers as a potential application for these ML methodologies is also interesting because the transport and biochemical reaction mechanisms of bio-towers are governed by advection-diffusion-reaction (ADR) equations. A protocol for approximating either the steady-state or unsteady state ADR equations of bio-towers by ML methods is also an unexplored realm in the literature. ADR equations are notoriously difficult to calculate within an optimization routine, so a protocol for its linear approximation can greatly advance the efficiency of computational routines in simulation and design software like Biowin, GPS-X, AQUASIM, SUMO, and SIMBA. In fact, none of this software currently have built-in process design optimization features, which could be attributed to the computational difficulty of calculating with the nonlinear biophysical models of biological wastewater systems for which they are based on. Given these gaps, the objectives of this study are:

- Investigate and compare existing ML algorithms for approximating nonlinear biophysical models of biological wastewater treatment systems.
- Develop a new method for generating a linear model that surpasses the ML algorithms mentioned above in terms of (i) computing time, (ii) prediction accuracy, and (iii) robustness when modeling both the activated sludge system (perfectly mixed, suspended growth system) and bio-tower system (advective-diffusive, attached growth system).
- Demonstrate the use of ML algorithms for optimal process design.

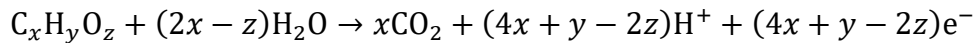
2 Theoretical Background

ammonia ($\text{NH}_3/\text{NH}_4^+$), nitrites (NO_2^-), and nitrates (NO_3^-), are crucial in wastewater treatment due to their light-independent energy generation and ability to thrive in environments with high carbon-to-nitrogen ratios and low light, unlike photosynthetic autotrophs. In wastewater treatment, autotrophs typically refer to these chemical autotrophs.

2.1 Organics Degradation

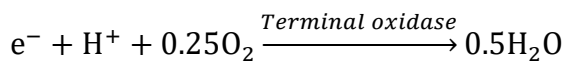
Heterotrophic energy metabolism entails the degradation of complex organic compounds ($\text{C}_x\text{H}_y\text{O}_z$), releasing CO_2 , protons (H^+), and energy. CO_2 is liberated as a gaseous byproduct, while H^+ is sequestered through association with a suitable acceptor in reduction-oxidation (redox) reactions. Aerobic wastewater treatment employs microorganisms that utilize O_2 as the terminal electron acceptor during the oxidation of organic compounds, a process known as aerobic respiration. The reaction equation representing this process are as follows (Xie, 2021):

Equation 2-1 Simplified reaction equation for aerobic respiration by heterotrophs in WWTPs (Xie, 2021)



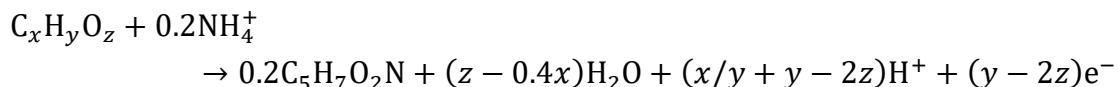
During aerobic respiration, $\text{C}_x\text{H}_y\text{O}_z$ serves as an electron donor and undergoes oxidation, as illustrated in **Equation 2-1**, while O_2 functions as an electron acceptor and is reduced by facilitating a terminal oxidase enzyme, as shown in **Equation 2-2**. The electron transfer generates a proton-motive force across the cell membrane, facilitating ATP synthesis.

Equation 2-2 Simplified reaction equation illustrating the electron transfer to O_2 catalyzed by a terminal oxidase enzyme in heterotrophs (Vandekerckhove, 2018)



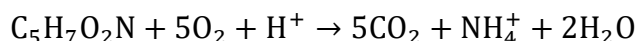
A fraction of $\text{C}_x\text{H}_y\text{O}_z$ is transformed into cell mass (biomass), as depicted in **Equation 2-3** (Vandekerckhove, 2018).

Equation 2-3 Simplified reaction equation for the conversion of $\text{C}_x\text{H}_y\text{O}_z$ into heterotrophic biomass (Vandekerckhove, 2018)



The empirical formula $\text{C}_5\text{H}_7\text{O}_2\text{N}$ typically represents heterotrophic biomass (Hoover & Porges, 1952). Endogenous respiration transpires under starvation conditions, with the simplified reaction equation provided in **Equation 2-4** (Takács & Vanrolleghem, 2006).

Equation 2-4 Simplified reaction equation for the endogenous respiration of heterotrophs (Takács & Vanrolleghem, 2006)



The energy released in these processes is proportional to the O_2 consumed, which permits the indirect measurement of carbonaceous energy release in aerobic biological systems. It is crucial to acknowledge that the precise stoichiometry of the aerobic respiration process and the amount of energy generated may vary depending on the specific organic compound involved.

2.2 Nitrification = Nitritation + Nitrataion

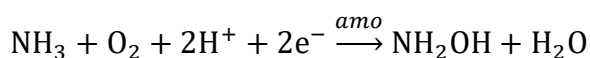
Contrasting heterotrophic processes, nitrifying microorganisms (i.e., chemical autotrophs) derive energy through the oxidation of inorganic N compounds, such as NH_4^+ and NO_2^- , during *nitrification*. Unlike heterotrophs, these autotrophs do not depend on organics for energy. Consequently, their O_2 demand is distinct from the O_2 demand associated with the degradation of organics by heterotrophs. Although O_2 demand experiments are not required to measure O_2 demand during nitrification, given the well-established stoichiometry of the nitrification process. The sequential reactions involving the oxidation of NH_4^+ to NO_2^- and subsequently to NO_3^- are detailed. The initial stage of nitrification, referred to as *nitritation*, involves the microbial-driven oxidation of NH_4^+ to NO_2^- . This conversion pathway is delineated in **Equation 2-5** to **Equation 2-8** (Preena et al., 2021; Van Kessel et al., 2015). Firstly, NH_4^+ dissociates to produce ammonia (NH_3) and a proton, as illustrated in **Equation 2-5** as follows:

Equation 2-5 Disassociation of NH_4^+ into NH_3 and H^+ (Preena et al., 2021; Van Kessel et al., 2015)



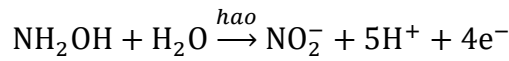
NH_3 is subsequently converted into hydroxylamine (NH_2OH) through the action of the membrane-bound enzyme ammonia monooxygenase (amo) in NH_3 -oxidizing bacteria (AOB), as follows:

Equation 2-6 Simplified reaction equation of the conversion of NH_3 into NH_2OH catalyzed by amo enzyme in AOB (Preena et al., 2021; Van Kessel et al., 2015)



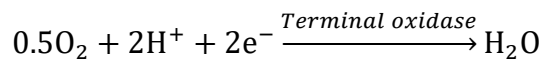
NH_2OH is then oxidized to NO_2^- , a reaction catalyzed by the periplasmic enzyme hydroxylamine oxidoreductase (hao), as follows:

Equation 2-7 Simplified reaction equation of the oxidation of NH_2OH to NO_2^- catalyzed by hao enzyme in AOB (Preena et al., 2021; Van Kessel et al., 2015)



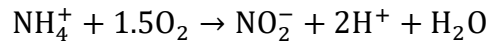
The electron released in this process is transferred to an electron transport chain, generating a proton-motive force that facilitates the synthesis of ATP. The electron transfer to O_2 is as follows:

Equation 2-8 Simplified reaction equation of the electron transfer to O_2 catalyzed by a terminal oxidase enzyme in nitrifying autotrophs (Preena et al., 2021; Van Kessel et al., 2015)



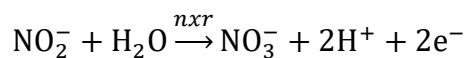
By combining the left- and right-hand sides of **Equation 2-5** to **Equation 2-7**, the total nitrification reaction by AOB is obtained (Van Kessel et al., 2015):

Equation 2-9 Simplified reaction equation of nitrification in AOB (Van Kessel et al., 2015)



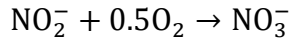
Nitrification represents the second step of the nitrification process wherein autotrophic NO_2^- -oxidizing bacteria (NOB) catalyze the oxidation of NO_2^- to NO_3^- (Preena et al., 2021). The majority of NOB can be categorized into several genera, including Nitrobacter, Nitrococcus, and Nitrotoga (phylum Proteobacteria), Nitrospira (phylum Nitrospirae), and Nitrospina (phylum Nitrospinae) (Preena et al., 2021). The enzyme nitrite oxidoreductase (nxr) facilitates the conversion of NO_2^- to NO_3^- (**Equation 2-10**), and the two electrons generated in this reaction are transferred to a terminal oxidase for respiratory purposes. The reaction equation for the electron transfer follows **Equation 2-8**. Some of the electrons produced are utilized to reduce CO_2 in the reductive acetyl-CoA pathway, which synthesizes autotrophic biomass. For brevity, the details of the reductive acetyl-CoA pathway are omitted here. The oxidation of NO_2^- to NO_3^- is depicted as follows:

Equation 2-10 Simplified reaction equation of the oxidation of NO_2^- into NO_3^- catalyzed by nxr enzyme in NOB (Preena et al., 2021)



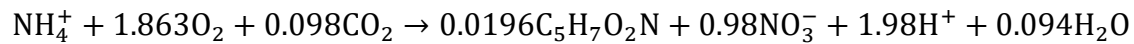
Combining **Equation 2-8** and **Equation 2-10** yields the total nitrification reaction by NOB (Van Kessel et al., 2015):

Equation 2-11 Simplified reaction equation of nitrification by NOB (Van Kessel et al., 2015)



The complete nitrification reaction incorporating autotrophic biomass yield is presented as follows:

Equation 2-12 Simplified reaction equation of the complete nitrification reaction by nitrifying autotrophs (Van Kessel et al., 2015)



2.3 Denitrification

Under anoxic conditions, heterotrophic microorganisms adapt their metabolic processes due to the absence of O_2 . Heterotrophic denitrifying bacteria utilize alternative electron acceptors, such as NO_3^- or NO_2^- , in a process known as denitrification. This process plays a pivotal role in N removal, converting NO_3^- or NO_2^- into nitrogen gas (N_2), which is released into the atmosphere. Heterotrophic denitrifying bacteria assimilate NO_3^- or NO_2^- from their environment and reduce them through a series of sequential steps. The heterotrophic denitrification process is generally contingent upon the electron donor involved. The common denitrification reactions occurring in biological wastewater treatment systems are presented in **Table 2-1**.

Table 2-1 Typical heterotrophic denitrification reactions in WWTPs

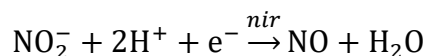
Electron donor	Reaction equation
Methanol	$1.08\text{CH}_3\text{OH} + \text{NO}_3^- + 0.24\text{H}_2\text{CO}_3 \rightarrow 0.056\text{C}_5\text{H}_7\text{O}_2\text{N} + 0.47\text{N}_2 + 1.68\text{H}_2\text{O} + \text{HCO}_3^-$
Ethanol	$0.69\text{C}_2\text{H}_5\text{OH} + \text{NO}_3^- + \text{H}^+ \rightarrow 0.14\text{C}_5\text{H}_7\text{O}_2\text{N} + 0.43\text{N}_2 + 2.07\text{H}_2\text{O} + 0.67\text{CO}_2$
Acetate	$0.846\text{CH}_3\text{COO}^- + \text{NO}_3^- \rightarrow 0.077\text{C}_5\text{H}_7\text{O}_2\text{N} + 0.462\text{N}_2 + 0.077\text{H}_2\text{O} + 1.308\text{CO}_2 + 1.846\text{OH}^-$
Glucose	$\text{C}_6\text{H}_{12}\text{O}_6 + 2.8\text{NO}_3^- + 0.5\text{NH}_4^+ + 2.3\text{H}^+ \rightarrow 0.5\text{C}_5\text{H}_7\text{O}_2\text{N} + 1.4\text{N}_2 + 6.4\text{H}_2\text{O} + 3.5\text{CO}_2$

2.4 Anammox

Certain autotrophic microorganisms utilize alternative electron acceptors during anaerobic conditions to facilitate denitrification. Anoxic NH_4^+ oxidation (anammox) involves the anaerobic oxidation of NH_4^+ using NO_2^- as an electron acceptor. Anammox bacteria, which are autotrophic, execute this process in three stages (Preena et al., 2021;

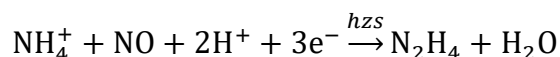
Van Kessel et al., 2015; Xie, 2021): Firstly, NO_2^- is reduced to nitric oxide (NO) by the nitrite reductase (nir) enzyme, as follows:

Equation 2-13 Simplified reaction equation of the reduction of NO_2^- to NO catalyzed by nir enzyme in anammox bacteria (Preena et al., 2021; Vandekerckhove, 2018)



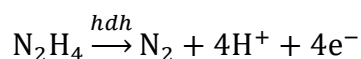
Secondly, NO combines with NH_4^+ to form hydrazine (N_2H_4) under the action of the hydrazine synthase enzyme (hzs), as follows:

Equation 2-14 Simplified reaction equation of the combination of NO with NH_4^+ to form N_2H_4 catalyzed by hzs enzyme in anammox bacteria (Dietl et al., 2015)



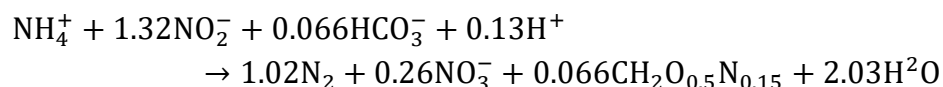
Third, the hydrazine dehydrogenase (hdh) enzyme catalyzes the oxidation of N_2H_4 to N_2 , as follows:

Equation 2-15 Simplified reaction equation of the oxidation of N_2H_4 to N_2 catalyzed by hdh enzyme in anammox bacteria (Maalcke et al., 2016)



The electrons produced in the process are transferred to an electron transport chain in **Equation 2-13** that induces a proton-motive force on the cell wall of the anammox bacteria, which is used to synthesize ATP. Additionally, during anammox anabolism, NO_2^- is oxidized to NO_3^- via the *nrx* enzyme (**Equation 2-10**), providing extra electrons for CO_2 fixation, which follows the reductive acetyl-CoA pathway (Xie, 2021). The details of the anabolic pathway are omitted here for brevity. The autotrophic biomass of anammox bacteria is represented by the empirical formula $\text{CH}_2\text{O}_{0.5}\text{N}_{0.15}$ (Strous et al., 1998). The complete anammox reaction incorporating autotrophic anammox biomass yield is presented as follows:

Equation 2-16 Simplified reaction equation of the complete anammox reaction by anammox bacteria (Strous et al., 1998)



In general, microorganisms metabolize contaminants (i.e., organics and NH_4^+) and transfer their electrons to electron acceptors such as O_2 , NO_3^- , or NO_2^- . In the process, they convert a portion of the catabolized contaminants into new biomass. This contributes

to the overall production of microbial biomass in the biological wastewater treatment system and the removal of contaminants from the wastewater. Consequently, the contaminant removal phenomena are subsequent to microbial growth phenomena, which makes the latter fundamental.

2.5 Biokinetic Modeling

Three things are apparent from the previous discussions. Firstly, multiple interrelated biochemical reactions are involved in the production and consumption of compounds in a biological wastewater treatment system. Secondly, these reactions are carried out by multiple organisms. Lastly, the production and consumption of compounds occur in fixed proportions. The ASM family of models were developed in the late 1990s in view of these facts (Barker & Dold, 1997; Dold et al., 1981; Gujer et al., 1999; Henze et al., 1987, 1999, 2006; Hu et al., 2007; Meijer, 2004; Nelson & Sidhu, 2009; Rieger et al., 2001). The reaction rate terms $R_i(\mathbf{C})$ in the ASM are interdependent and are governed by the stoichiometric relationships. The stoichiometric matrix $\mathbf{S}$ encapsulates the fixed proportions of consumption and production of compounds during biochemical reactions (Henze et al., 2006). Specifically, the relationship between the reaction rate vector $\mathbf{R}$ and the biokinetic process vector $\rho(\mathbf{C})$ is given by:

$$\mathbf{R} = \mathbf{S} \cdot \rho(\mathbf{C})$$

where:

- $\mathbf{R} = [R_1, R_2, \dots, R_n]$ is the metabolic rate vector for n compounds,
- $\mathbf{S} = [S_{i,j}]$ is the stoichiometric matrix where $S_{i,j}$ represents the stoichiometric coefficient of compound i in reaction j ,
- $\rho(\mathbf{C}) = [\rho_1(\mathbf{C}), \rho_2(\mathbf{C}), \dots, \rho_m(\mathbf{C})]$ is the biokinetic process vector for m reactions,

Table 2-2 Sample tabular representation of the relationships of metabolic rate, stoichiometric coefficients, and biokinetic processes

Biokinetic processes ↓	Compounds				
	C_1	C_2	...	C_i	...
$\rho_1(\mathbf{C})$	$S_{1,1}$	$S_{1,2}$	...	$S_{1,i}$	...
$\rho_2(\mathbf{C})$	$S_{2,1}$	$S_{2,2}$	...	$S_{2,i}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	...
$\rho_j(\mathbf{C})$	$S_{j,1}$	$S_{j,2}$	...	$S_{j,i}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
Metabolic rate →	R_1	R_2	...	R_i	...

A tabular representation of the relationship of metabolic rates, stoichiometric coefficients, and biokinetic processes is presented in **Table 2-2**. The stoichiometric matrix $\mathbf{S}$ plays a crucial role in linking the consumption and production rates of different compounds within

the system. It should be noted that the coefficients in the stoichiometric equations presented in the previous discussions are the elements of this matrix. Each element $S_{i,j}$ of the matrix ensures that the conservation of mass is maintained during biochemical transformations. For example, in the context of microbial growth and pollutant degradation, the stoichiometric matrix captures the fixed ratios in which substrates are consumed, and biomass is produced or maintained. The nonlinearity in ASM equations arises primarily from the biokinetic rates $\rho(\mathbf{C})$, which typically involve nonlinear functions of $\mathbf{C}$. For instance, microbial growth rates often follow Monod kinetics, which are nonlinear with respect to substrate concentration:

$$\mu_j(\mathbf{C}) = \mu_{max,j} \prod_{i=1}^m \frac{C_i}{K_{j,i} + C_i}$$

where:

- $\mu_j(\mathbf{C})$ is the specific growth rate of microorganisms involved in process j and is an element in $\rho(\mathbf{C})$,
- $\mu_{max,j}$ is the maximum specific growth rate for process j ,
- $K_{j,i}$ is the half-saturation constant for substrate i in process j ,
- m is the number of compounds involved in process j .

There are ways to linearize the reaction rate vector $\mathbf{R}$. The most common approach is the use Taylor series expansion (M. Li et al., 2020; T. Yang et al., 2014), which is fine for optimal control applications, but this approach greatly limits the search space for design optimization applications because the model is only valid around vicinity of the approximation point. This limitation can be addressed by taking advantage of the structure of the Monod function. Essentially, $\mu_j(\mathbf{C})$ is a product of a constant $\mu_{max,j}$ and the multiplication of some function of C_i for all i . This implies that all compounds are correlated to each other by virtue of $\mathbf{R}$ being a vector of compound production or consumption. There are a multitude of ways to elucidate these correlations, but the important aspect of this realization is that it allows us to lump the effects of stoichiometric coefficients to the production and consumption of compounds into correlation coefficients in some empirical model. To this end, ML models can be used, which is the premise of this study. In fact, several studies have deployed ML techniques to approximate nonlinearities in biological wastewater treatment process (Ahn et al., 2014; Asadi et al., 2017; Fang et al., 2011; Kim et al., 2009; Kusiak & Wei, 2012; Lim et al., 2011; Wu et al., 2016), which substantiates the theoretical arguments posed here, but these existing approaches are not without gaps, which are highlighted in the next section.

3 Review Related Literature

3.1 Foundational Activated Sludge Models

The Activated Sludge Model (ASM) family represents one of the most significant developments in modern wastewater treatment engineering, providing a standardized framework for simulating and designing biological treatment processes in activated sludge systems (Henze et al., 2006). Initially developed under the aegis of the International Association on Water Quality (IAWQ), which eventually became the International Water Association (IWA), these models consolidated diverse research efforts and practical experiences into a unified, mathematical representation of critical microbial and chemical reactions (Henze et al., 2006). The need for such standardized models arose due to the inherent complexity of biological wastewater treatment: multiple microbial populations compete and cooperate to remove carbon, nitrogen, and, eventually, phosphorus from the influent stream (Dold et al., 1981). By creating a common platform for predicting effluent quality and guiding design modifications, the ASM family fundamentally transformed both scientific inquiry and operational decision-making in wastewater engineering (citation needed). (Dold et al., 1981; Henze et al., 1999; Nelson & Sidhu, 2009; Vanrolleghem et al., 1999)

ASM1 was the first model in this series, introduced in 1987 by Henze, Grady, Gujer, Marais, and Matsuo (Henze et al., 1987; Van Loosdrecht et al., 2015). It focused primarily on carbon oxidation and nitrogen removal processes, capturing the essential dynamics of heterotrophic and autotrophic biomass (Van Loosdrecht et al., 2015). Within ASM1, heterotrophs metabolize readily biodegradable substrate and oxygen under aerobic conditions, while they denitrify under anoxic conditions by using nitrate as an electron acceptor (Henze et al., 2006). Autotrophs, on the other hand, nitrify ammonia to nitrite and then to nitrate, representing one of the most critical steps in maintaining nitrogen balance in effluents (Henze et al., 1987, 2006). The ability of ASM1 to predict effluent quality for key parameters—such as chemical oxygen demand (COD), biochemical oxygen demand (BOD), ammonia, and nitrate—rendered it a powerful tool for both researchers and plant operators, even though the model did not explicitly address phosphorus removal (Henze et al., 1999). Despite its limitations, ASM1's relatively straightforward approach to substrate utilization, biomass growth, and decay established the core framework still evident in later models (Henze et al., 2006; Van Loosdrecht et al., 2015).

Building on this foundation, ASM2 emerged in the mid-1990s to address enhanced biological phosphorus removal (EBPR) processes by introducing phosphorus-accumulating organisms (PAOs) (Henze et al., 1999). This expanded scope reflected the growing interest among engineers and regulators in controlling eutrophication, a process driven by excessive nutrient discharge into natural water bodies (Selerio, 2022). Under

ASM2, PAOs absorb phosphate during aerobic conditions and release it in anaerobic phases, a dynamic that significantly improves phosphorus removal efficiency (Henze et al., 1999, 2006). Though ASM2 retained the core processes of heterotrophic and autotrophic metabolism from ASM1, it provided additional state variables and kinetic expressions to capture the interplay between carbon, nitrogen, and phosphorus transformations. In subsequent refinements, ASM2d added the capacity for PAOs to denitrify (i.e., use nitrate as an electron acceptor) under anoxic conditions, further integrating nitrogen and phosphorus removal pathways (Henze et al., 1999). While the inclusion of these processes offered a more comprehensive outlook, the expanded parameter set demanded more extensive calibration data, which sometimes posed challenges for real-world applications (Petersen et al., 2003; Vanrolleghem et al., 1999).

Meanwhile, ASM3, introduced around 2000, sought to streamline certain aspects of ASM1 and ASM2 by revising the way hydrolysis and biomass growth were modeled (Gujer et al., 1999). This model focused primarily on carbon and nitrogen removal, returning to a simpler structure for plants not dealing with intense phosphorus standards or where EBPR was less of a priority (Gujer et al., 1999). Specifically, ASM3 decoupled the hydrolysis process from direct heterotrophic growth, allowing a more detailed representation of how particulate organic matter is converted into soluble substrate (Gujer et al., 1999; Henze et al., 2006). These refinements facilitated more accurate simulations of sludge production and effluent quality under varying operational conditions, such as fluctuating influent loads, changing temperatures, and intermittent aeration (Henze et al., 2006). Still, ASM3 did not natively incorporate EBPR, meaning that engineers prioritizing phosphorus removal often reverted to ASM2 or ASM2d (Hauduc et al., 2013; Henze et al., 2006).

Across all ASM variants, the primary contributions to wastewater treatment design and optimization include the ability to predict effluent quality under different operating regimes, test hypothetical configurations before full-scale implementation, and guide the selection of key parameters such as sludge retention time, aeration intensity, and reactor geometry (Wang et al., 2024). Calibrating these models, however, requires extensive data on wastewater characteristics, microbial community composition, and environmental conditions, leading to potential challenges in practice (Petersen et al., 2003). Furthermore, non-linear terms—such as Monod kinetics and complex stoichiometric relationships—can pose computational hurdles, especially when operators or engineers attempt to integrate ASM simulations with exact or heuristic optimization techniques (M. Li et al., 2020; T. Yang et al., 2014). Nevertheless, the ASM family remains a cornerstone of activated sludge research and design.

3.2 ASM Extensions and Comparisons

A comparison of known ASM models to date is presented in **Table 3-1**,

Table 3-2, and

Table 3-3. The Activated Sludge Models (ASMs) included in these tables represent both the foundational ASM family discussed above—namely ASM1, ASM2d, and ASM3—and several notable extensions or alternatives such as Barker & Dold (1997), ASM3 + BioP (Rieger et al., 2001), UCTPHO+ (Hu et al., 2007), and ASM2d + TUD (Meijer, 2004). These models differ in their inclusion of certain processes (carbon, nitrogen, and phosphorus removal), in the number and type of state variables they track, and in the stoichiometric and kinetic parameters they use to describe the microbial conversions within wastewater treatment systems.

The **Table 3-1** provides a comparative overview of the general characteristics for each model, including the type of substrates (C, N, or P) considered, the number of processes, the number of state variables, and the size and complexity of the interaction matrix. Models such as ASM1 (Van Loosdrecht et al., 2015) and ASM3 (Gujer et al., 1999) focus primarily on carbon and nitrogen and, as a result, have fewer processes and state variables. In contrast, Barker & Dold (1997), ASM2d (Henze et al., 1999), ASM3 + BioP (Rieger et al., 2001), UCTPHO+ (Hu et al., 2007), and ASM2d + TUD (Meijer, 2004) include phosphorus removal via enhanced biological phosphorus removal (EBPR), which significantly increases the total number of processes and state variables. The inclusion of PAOs (phosphate-accumulating organisms) typically leads to additional complexity since both carbon and phosphorus transformations must be accounted for under varying aerobic, anoxic, and anaerobic conditions. This inclusion not only increases the number of interactions between processes and variables but also expands the stoichiometric composition matrix and the number of temperature adjustments required to reflect different kinetic rates under different thermal regimes.

The

Table 3-2 highlights the stoichiometric parameters underlying these models. Each row reveals how many distinct stoichiometric coefficients or yield-related parameters are assigned to key process groups such as hydrolysis, ordinary heterotrophic organisms (OHO), autotrophic nitrifiers (ANO), and phosphate-accumulating organisms (PAO). Models like ASM1, which provides only one stoichiometric parameter each for OHO and ANO (and excludes PAOs), appear relatively simple (Henze et al., 2006). Barker & Dold (1997) or ASM2d + TUD (Meijer, 2004), on the other hand, distribute many stoichiometric parameters among hydrolysis, heterotrophs, nitrifiers, and PAOs, offering a more granular portrayal of the biochemical transformations for EBPR. A larger number of stoichiometric parameters for PAOs (as high as 12 in ASM2d + TUD) indicates a detailed representation of intracellular storage compounds such as poly-P, glycogen, or PHA/PHB, further revealing the complexity of phosphorus uptake and release processes. In some instances, the models provide a single “biomass general” stoichiometric parameter for

decay or maintenance, whereas others prefer to split such considerations across multiple functional groups (Meijer, 2004).

The

Table 3-3 compares kinetic parameters, summarizing the rate constants and half-saturation coefficients for hydrolysis, heterotrophic growth, nitrification, and PAO activities. Certain models, like ASM2d, distribute many kinetic parameters (e.g., six for hydrolysis, twelve for OHO, and eighteen for PAO), reflecting multiple considerations such as different hydrolysis pathways (particulate to soluble under aerobic, anoxic, or anaerobic conditions) or different PAO growth and storage phases (Henze et al., 1999, 2006). ASM3 separates heterotrophic biomass growth from hydrolysis in a different manner, resulting in fewer hydrolysis parameters but more for heterotrophic conversion. Where EBPR is included, particularly in Barker & Dold, ASM2d + TUD, ASM3 + BioP, or UCTPHO+, the number of PAO-related kinetic parameters escalates. This arises from the need to describe multiple, phase-specific processes: anaerobic substrate uptake and storage, anoxic or aerobic storage mobilization, phosphate uptake or release, and interactions with nitrification or denitrification (Henze et al., 2006). Some models include one general kinetic term for biomass decay, whereas others specify separate decay terms for each major functional group.

Across these three tables, it is evident that the main difference between older, simpler models such as ASM1 or ASM3 and more comprehensive models like Barker & Dold, ASM2d, ASM3 + BioP, UCTPHO+, or ASM2d + TUD lies in the extent to which phosphorus removal and detailed EBPR mechanisms are incorporated. The original ASM1 described only organic carbon removal and nitrogen conversions with a limited set of stoichiometric and kinetic parameters. ASM2d then added EBPR features to account for PAOs, refining many stoichiometric and kinetic relationships to match observed performance under anaerobic–anoxic–aerobic conditions. ASM3 redefined heterotrophic biomass in a more dynamic manner, simplifying hydrolysis yet expanding certain kinetic aspects, particularly for the heterotrophs. Subsequent extensions such as ASM3 + BioP or UCTPHO+ introduce dedicated mechanisms for PAOs and sometimes GAOs (glycogen-accumulating organisms), expanding the parameter set considerably.

Overall, the growing complexity of these models provides more nuanced predictions of real wastewater treatment processes, especially in systems aiming to meet increasingly stringent phosphorus (and sometimes simultaneous nitrogen–phosphorus) effluent standards. However, more detailed models require correspondingly more effort in calibration and validation, as they involve numerous processes and organism groups. The choice of model is therefore influenced by the treatment objectives, available operational and analytical data, the extent of EBPR, and the level of detail required for process optimization or research. In practice, practitioners often begin with simpler models like ASM1 or ASM3 for broad design guidance (Hauduc et al., 2013) and switch to more

sophisticated models—such as Barker & Dold, ASM2d, UCTPHO+, or ASM2d + TUD—for advanced phosphorus control, specialized operational strategies, or detailed research into the metabolic pathways of different microbial communities.

Table 3-1 Comparison of activated sludge models: general parameters

Models	Refs	Substrates	# of processes	# of state variables	# of interacting processes vs variables	Total	Composition matrix parameters	Temperature adjustment
ASM1	Henze et al. (2006)	CN	8	13	31	26	2	7
Barker & Dold	Barker & Dold (1997)	CNP	36	19	153	81	16	18
ASM2d	Henze et al. (1999)	CNP	21	19	136	74	13	13
ASM3	Gujer et al. (1999)	CN	12	13	72	46	8	10
ASM3 + BioP	Rieger et al. (2001)	CNP	23	17	148	83	15	13
UCTPHO+	Hu et al. (2007)	CNP	35	16	169	66	12	10
ASM2d + TUD	Meijer (2004)	CNP	22	18	154	98	16	15

Table 3-2 Comparison of activated sludge models: stoichiometric parameters

Models	Refs	Stoichiometry - Hydrolysis	Stoichiometry - OHO	Stoichiometry - ANO	Stoichiometry - PAO	Stoichiometry - Biomass general
ASM1	Henze et al. (2006)	-	1	1	-	1
Barker & Dold	Barker & Dold (1997)	2	5	2	8	-
ASM2d	Henze et al. (1999)	1	1	1	3	1
ASM3	Gujer et al. (1999)	1	4	1	-	1
ASM3 + BioP	Rieger et al. (2001)	1	4	1	5	1
UCTPHO+	Hu et al. (2007)	-	3	2	7	-
ASM2d + TUD	Meijer (2004)	1	2	2	12	-

Table 3-3 Comparison of activated sludge models: kinetic parameters

Models	Refs	Kinetic Hydrolysis	- Kinetic OHO	- Kinetic - ANO	Kinetic - PAO	Kinetic Biomass general	-
ASM1	Henze et al. (2006)	3	6	5	-	-	-

Barker & Dold	Barker & Dold (1997)	4	9	5	11	1
ASM2d	Henze et al. (1999)	6	12	6	18	-
ASM3	Gujer et al. (1999)	2	13	6	-	-
ASM3 BioP	+ Rieger et al. (2001)	2	13	7	21	-
UCTPHO+	Hu et al. (2007)	-	13	4	14	1
ASM2d TUD	+ Meijer (2004)	6	12	6	26	-

3.3 Advection-Diffusion-Reaction (ADR) Models

While the ASM family of models were originally used for modeling the activated sludge process, which is a suspended-growth system, it has also been widely adopted for modeling attached-growth systems such as biofilm technologies (Hellal & Abou-Elela, 2021; Lainé et al., 1999; Liang et al., 2021; Simsek et al., 2012). Bio-towers, sometimes referred to as trickling filters, are among the most widely used attached-growth systems for biological wastewater treatment and are particularly effective for removing organic and nitrogenous pollutants from domestic and industrial effluents (Mahmoud et al., 2011, 2018; Tawfik et al., 2006, 2009; Tyagi et al., 2021). In these systems, a microbial biofilm develops on a packed medium (e.g., rocks, structured plastic media) while wastewater percolates through, enabling microorganisms to metabolize dissolved pollutants under aerobic or anoxic conditions (Mahmoud et al., 2010, 2011; Nasr et al., 2022). Bio-towers are generally appreciated for their relatively low operational costs and ability to handle fluctuations in wastewater loading, although their design and performance optimization can be complex due to the interplay between hydrodynamics, mass transfer, and biological reactions (Bressani-Ribeiro et al., 2021; Nasr et al., 2022; Tyagi et al., 2021). Accurately modeling these phenomena requires a rigorous description of fluid movement, solute dispersion, and microbial transformations, which has led many researchers to adopt Advection-Diffusion-Reaction (ADR) equations as the fundamental modeling framework (Sadino-Riquelme et al., 2023).

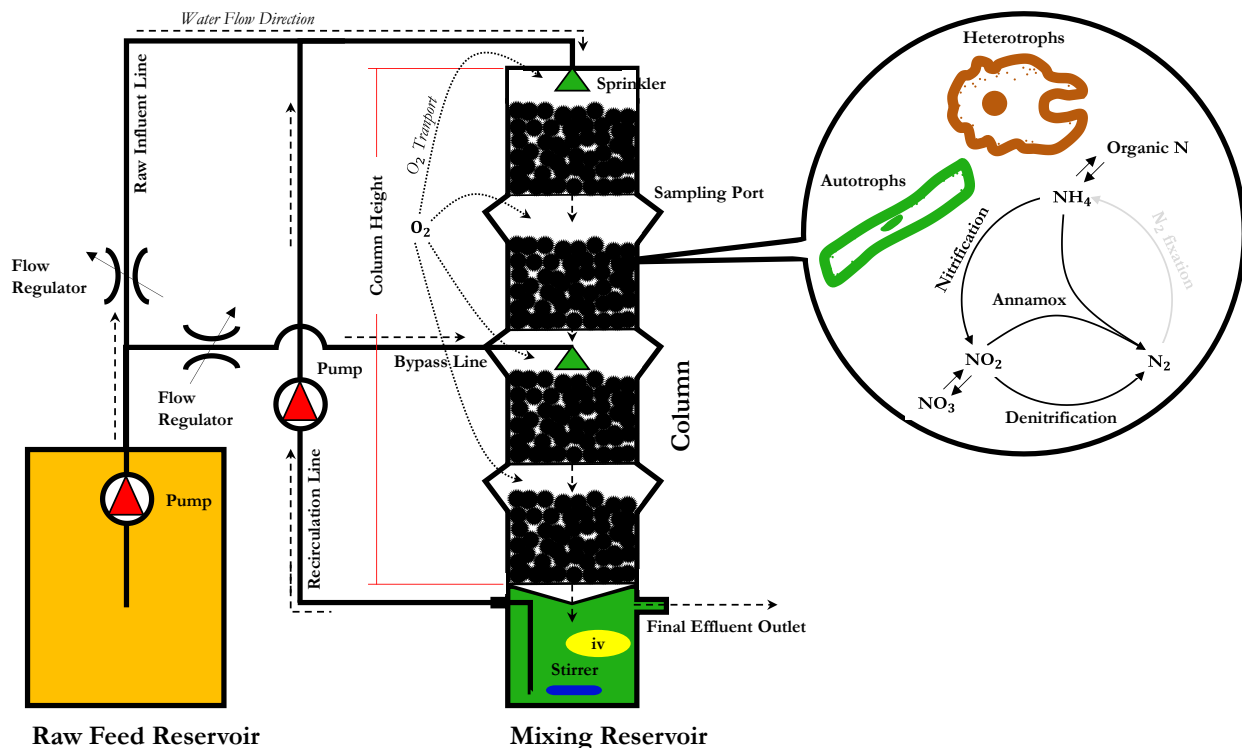


Figure 3-1 Passively aerated bio-tower

A passively aerated bio-tower and its operation is illustrated in **Figure 3-1**. In this sample setup, the wastewater is first pumped into the bio-tower, which elevates the fluid to a specific column height. As it enters the system, it passes through a sprayer, which distributes the wastewater evenly across the column. The bulk wastewater flows downward through the column and is exposed to the biological layers within the bio-tower. The treatment cycle that occurs in the system includes nitrification, an aerobic reaction where ammonia (NH_4^+) is converted to nitrite (NO_2^-) and subsequently to nitrate (NO_3^-). This step is facilitated by nitrifying bacteria that thrive in oxygen-rich environments. As the nitrified effluent moves through the system, it encounters anoxic zones where anammox (anaerobic ammonium oxidation) bacteria are present. The anammox process enables the conversion of ammonium and nitrite into nitrogen gas (N_2) without the need for oxygen. This step is particularly effective in reducing nitrogen compounds and is essential for minimizing nitrogen pollution. Following the anammox process, any remaining nitrate and nitrite undergo denitrification in an anaerobic environment. Here, heterotrophic denitrifying bacteria utilize the remaining nitrogen oxides (NO_2^- and NO_3^-) as electron acceptors and converting them into nitrogen gas (N_2), which is then released harmlessly into the atmosphere. Additionally, organic nitrogen compounds are broken down by heterotrophic bacteria through the process known as ammonification, which convert them to ammonia.

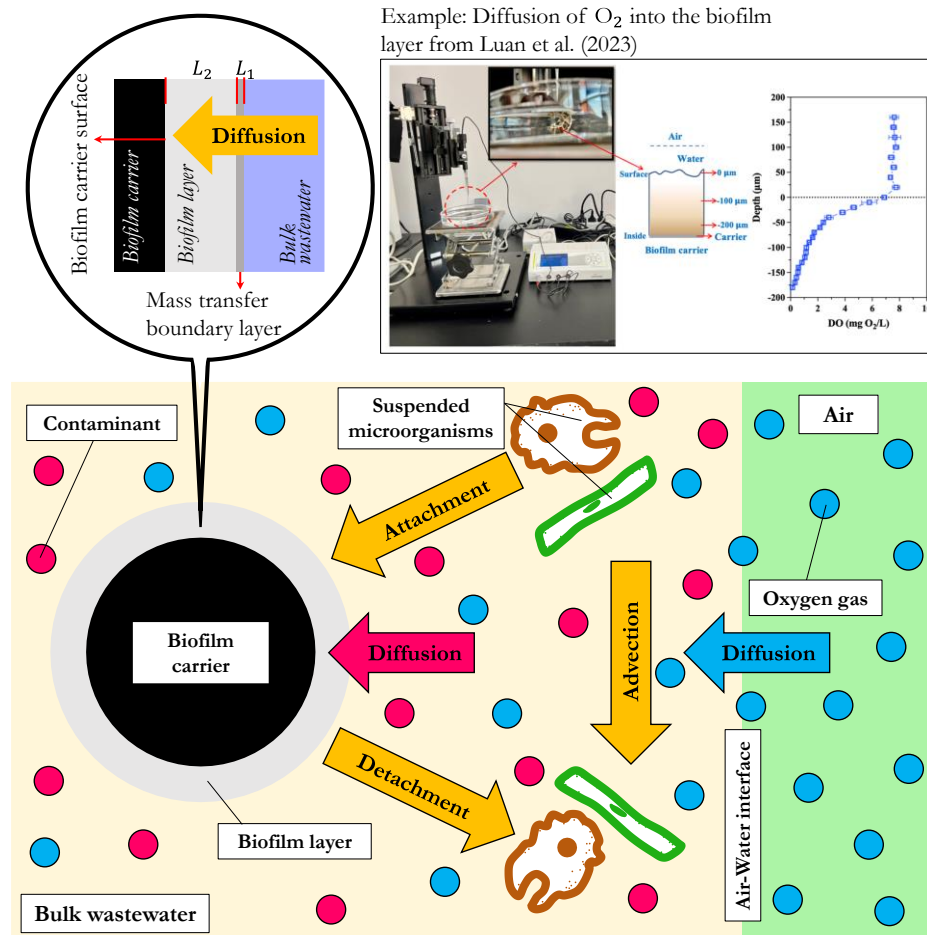


Figure 3-2 Transport mechanisms in a bio-tower

In a bio-tower, several transport mechanisms facilitate the movement of contaminants, nutrients, and gases within the system (Luan et al., 2023), which are illustrated in **Figure 3-2**. Diffusion allows oxygen and contaminants in bulk wastewater to enter the biofilm layer. They move from high concentration areas in the bulk or air interface to lower concentrations within the biofilm, which supply microorganisms with essential resources for metabolism. At the same time, advection brings bulk wastewater and suspended microorganisms closer to the biofilm carrier that facilitates contact. Microorganisms in the bulk water then attach to the biofilm layer. As the biofilm thickens over time, parts of it may detach, where microorganisms or biofilm fragments break off and return to the bulk water. This detachment helps maintain biofilm density.

The ADR approach offers a way to capture both the transport of compounds in the flowing wastewater and the biological or chemical transformations taking place within the biofilm and bulk fluid (Bressani-Ribeiro et al., 2021). In its simplest one-dimensional form, the ADR equation for a single dissolved substrate C can be expressed as follows:

$$\frac{\partial C}{\partial t} = -v \frac{\partial C}{\partial z} + D \frac{\partial^2 C}{\partial z^2} + R(C, X)$$

This formulation assumes that variations occur primarily along the vertical axis z of the filter (from top to bottom), while lateral variations are neglected or approximated with an effective dispersion term (Wik, 2003). Researchers often expand this model to multiple dimensions or multiple coupled species (such as ammonia, nitrite, nitrate, and oxygen), thereby increasing the realism but also the computational burden (Hellal & Abou-Elela, 2021; Lainé et al., 1999; Liang et al., 2021; Simsek et al., 2012). In the equation, $\frac{\partial C}{\partial t}$ represents the rate of change of the substrate concentration over time, which is influenced by three main factors: advection $-v \frac{\partial C}{\partial z}$, diffusion $D \frac{\partial^2 C}{\partial z^2}$, and biological or chemical reaction $R(C, X)$ (Díez-Montero et al., 2019; Kasi et al., 2011; Liang et al., 2021; Wik, 2003). The term v is the advective velocity, describing how quickly the wastewater flows downward through the bio-tower, and D is an effective diffusion or dispersion coefficient that accounts for both molecular diffusion and small-scale eddy dispersion in the liquid film (Sadino-Riquelme et al., 2023). The reaction rate $R(C, X)$ typically incorporates non-linear kinetic expressions (e.g., Monod-type equations) that depend on the local substrate concentration C and microbial special concentration X (Díez-Montero et al., 2019; Kasi et al., 2011; Liang et al., 2021; Wik, 2003). The reaction rate $R(C, X)$ is typically modeled using any of the ASM family of models discussed previously (Bressani-Ribeiro et al., 2021). Additional coupled PDEs are often required for other species, including dissolved oxygen and nitrogen compounds, resulting in a highly interdependent system of equations that must be solved simultaneously (Sadino-Riquelme et al., 2023).

Modeling bio-towers with ADR equations becomes even more complex when one considers the heterogeneous flow distribution, dynamic loading conditions, and variable microbial community structures within the biofilm (Sadino-Riquelme et al., 2023). In reality, wastewater may not uniformly trickle downward; some regions of the packed medium receive more flow than others, creating non-ideal flow regimes (Wik, 2003). The biofilm itself can vary in thickness and porosity over time, affecting diffusion pathways and reaction rates (T. C. Zhang & Bishop, 1994). Furthermore, the microbial composition changes in response to nutrient availability and oxygen diffusion, introducing additional uncertainty and temporal variability into the model (Hatamoto et al., 2018; Lu et al., 2014; D. Zhang et al., 2016). These complexities often lead to stiff and computationally expensive numerical simulations, prompting the development of specialized software—such as AQUASIM, GPS-X, Biowin, SUMO, and SIMBA—to tackle ADR-based modeling for real-world systems (Bazemo et al., 2021; Caro et al., 2024; Díez-Montero et al., 2019; Hellal & Abou-Elela, 2021; Jafarinejad, 2020; Jasim, 2020; Kasi et al., 2011; Lainé et al., 1999; Liang et al., 2021; Nazif et al., 2023; Sadri Moghaddam & Pirali, 2021; Simsek et al., 2012).

Despite the accuracy that detailed ADR models can provide, practical engineering applications frequently require iterative trial-and-error design or heuristic optimization approaches due to the high computational demands of solving large systems of coupled partial differential equations (Sadino-Riquelme et al., 2023; Wang et al., 2024). When the objective is to optimize bio-tower performance—whether by adjusting hydraulic loading rates, selecting media configurations, or establishing control strategies—repeated ADR simulations can be prohibitively time-intensive (Croll et al., 2023). Hence, researchers have explored surrogate modeling techniques, including artificial intelligence and machine learning, to approximate the results of full ADR models while reducing computational load (Ching et al., 2022; Croll et al., 2023; Ekinici et al., 2023; Sundui et al., 2021; Waqas et al., 2022). These surrogate models can guide system designers toward near-optimal configurations in less time, thus bridging the gap between theoretical accuracy and real-world feasibility (Sundui et al., 2021).

3.4 Software Packages for Modeling and Simulation

A range of specialized software has been developed for modeling and simulating biological wastewater treatment processes, enabling engineers and researchers to study treatment performance under a variety of operating conditions. Among the most commonly used packages are GPS-X, Biowin, AQUASIM, SUMO, and SIMBA (Bazemo et al., 2021; Caro et al., 2024; Díez-Montero et al., 2019; Hellal & Abou-Elala, 2021; Jafarinejad, 2020; Jasim, 2020; Kasi et al., 2011; Lainé et al., 1999; Liang et al., 2021; Nazif et al., 2023; Sadri Moghaddam & Pirali, 2021; Simsek et al., 2012). Each of these platforms allows users to construct detailed representations of wastewater treatment processes—often including activated sludge systems, fixed-film reactors (like bio-towers), primary and secondary clarifiers, and additional unit operations. By offering pre-configured templates of common processes, comprehensive libraries of kinetic models, and graphical user interfaces, these software packages significantly reduce the effort and expertise needed to set up sophisticated simulations [citation needed]. They typically include modules for dynamic simulations, parameter estimation, and calibration, enabling users to capture diurnal load variations, investigate chemical dosing strategies, or test the impact of changes in aeration rates and sludge retention times.

GPS-X, developed by Hydromantis, is known for its intuitive interface and modular approach, allowing users to drag and drop unit processes to build a virtual treatment plant (Sadri Moghaddam & Pirali, 2021). It offers a variety of standard and advanced biological models—such as ASM1, ASM2d, ASM3, and extensions for nitrification, denitrification, and phosphorus removal—along with functionalities for process control testing and scenario analysis (Jasim, 2020). Biowin, created by EnviroSim, similarly provides comprehensive modeling features and is lauded for its user-friendly calibration routines that guide practitioners in matching model predictions to real plant data (Caro et al., 2024; Díez-Montero et al., 2019; Kasi et al., 2011; Simsek et al., 2012). AQUASIM, meanwhile,

was initially developed at Eawag (Swiss Federal Institute of Aquatic Science and Technology) with a particular focus on controlling and analyzing aquatic systems; its open-ended structure is conducive to customizing differential equations, making it especially popular among researchers exploring unique or novel reactor configurations (Bressani-Ribeiro et al., 2021). SUMO, by Dynamita, and SIMBA, by IFAK, are also prominent: they incorporate models for activated sludge, biofilm reactors, and advanced biological nutrient removal, frequently coupling these with optional extensions for anaerobic digestion and side-stream processes (Bazemo et al., 2021; Jafarinejad, 2020; Lainé et al., 1999; Liang et al., 2021; Nazif et al., 2023).

Despite the extensive functionalities of these platforms, embedded optimization capabilities—particularly near-global optimization or advanced control—remain notably absent or underdeveloped in most packages. The primary focus of these software environments is on simulation, calibration, and scenario analysis; typically, engineers will tweak parameters iteratively (often referred to as a “trial-and-error” approach) to find reasonable solutions that meet regulatory standards (Croll et al., 2023). This manual or semi-manual procedure can be sufficient when the primary aim is to ensure compliance or gain insights into possible operational strategies. However, for formal optimization—such as minimizing energy costs, maximizing treatment efficiency, or balancing multiple objectives—users would need to embed these simulation tools inside an external optimization framework, which can be cumbersome and computationally demanding (Caro et al., 2024).

A reason for this gap is the computational complexity associated with wastewater treatment models. Biological and physico-chemical processes in reactors like activated sludge systems or bio-towers are inherently non-linear, often described using multiple, interlinked partial differential equations (e.g., for advection-diffusion-reaction modeling) that must be solved simultaneously (Díez-Montero et al., 2019; Kasi et al., 2011; Liang et al., 2021; Sadino-Riquelme et al., 2023; Wik, 2003). These equations can be numerically stiff and time-consuming to simulate, especially if the models include multiple state variables (e.g., COD, BOD, nitrogen, phosphorus, and different microbial populations) and dynamic conditions (like hourly fluctuations in influent loads) (M. Li et al., 2020; Wik, 2003; T. Yang et al., 2014). The software would have to iteratively run many simulations—sometimes in the thousands, millions, or even billions—to fully explore the decision space and converge on near-optimal solutions. Given these challenges, developers of packages like GPS-X, Biowin, AQUASIM, SUMO, and SIMBA have, to date, concentrated on making the simulation aspect robust and user-friendly rather than incorporating complex built-in optimization engines that would risk long run times and limited performance on standard computers.

Nevertheless, research groups and advanced users often couple these simulation tools with external optimization frameworks—for instance, by writing scripts in platforms such

as MATLAB, Python, or specialized optimization software—to systematically evaluate thousands of simulation scenarios (Caro et al., 2024). While this workflow can yield powerful results, it typically demands significant computational power (e.g., high-performance clusters) and specialized expertise in both modeling and optimization. The lack of a direct, integrated optimization module within mainstream wastewater simulation software remains a known limitation, especially for practitioners seeking straightforward, single-click optimization solutions. Looking ahead, there is growing interest in surrogate modeling or machine learning (ML) methods that can approximate complex simulation outputs and thereby drastically reduce the computational burden, making embedded or near-real-time optimization more feasible (J. Li et al., 2022; Wang et al., 2024; Waqas et al., 2022). If such surrogate or reduced-order models become more common, they may be integrated into future releases of these software packages, potentially closing the gap between simulation and embedded optimization in the wastewater treatment domain.

3.5 Optimization Problems

In the context of biological wastewater treatment, optimization problems typically involve multiple, and often conflicting, goals: minimizing costs, conserving resources, and ensuring compliance with regulatory standards for effluent quality. These objectives must be addressed within the confines of practical limitations such as finite source capacities, operational or technological constraints of treatment units, and uncertainties in pollutant loads. Furthermore, diverse system configurations—from integrated water and wastewater networks with various mixing and routing schemes to complex multi-stage treatment plants—add layers of complexity to the decision-making process. By leveraging optimization models, engineers and planners can systematically evaluate feasible system configurations, assign process flows, and fine-tune operational parameters to identify the most effective solutions. Three general optimization problems arise in wastewater treatment literature, which are discussed here.

3.5.1 Water/Wastewater Reuse Network Design Problem

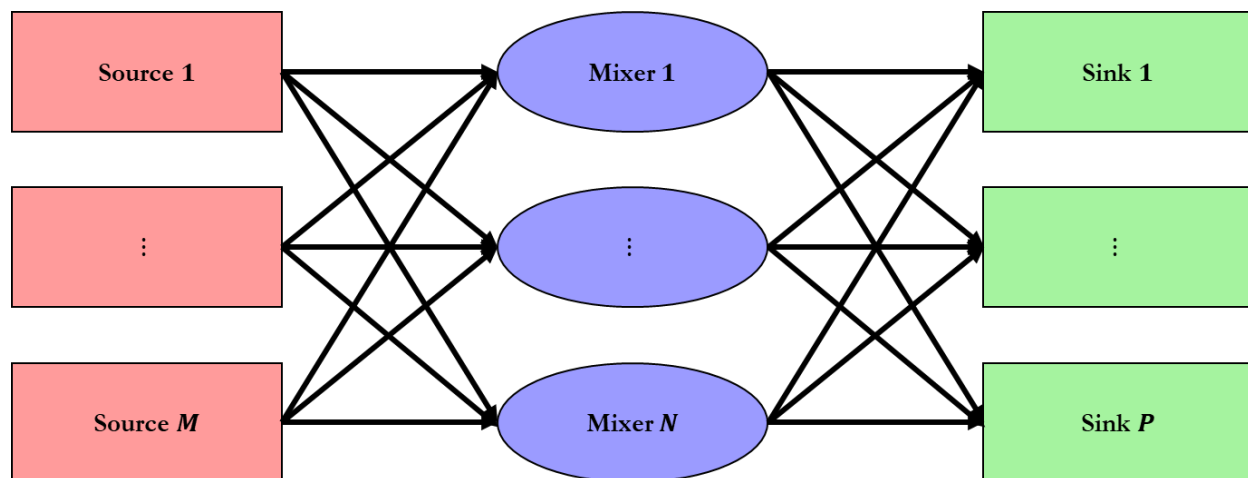


Figure 3-3 Water/wastewater reuse network design problem superstructure concerning multiple sources, mixers, and sinks.

The first problem addresses the routing and allocation of water and wastewater through a network of sources, mixers, and sinks. A common challenge in wastewater management arises when designing and operating a network consisting of multiple water sources, intermediate mixers, and sinks (Hansen et al., 2018). This problem is widely tackled in the water/wastewater reuse optimization literature (Guelli Ulson De Souza et al., 2011; Gutterres et al., 2010; Hansen et al., 2018; Rubio-Castro et al., 2013; Xu et al., 2015). A general source-mixer-sink presentation* of the problem is shown in **Figure 3-3** and its mathematical formulation is as follows:

Here, we assume there are M sources (denoted Source 1, Source 2, ..., Source M), N mixers (Mixer 1, Mixer 2, ..., Mixer N), and P sinks (Sink 1, Sink 2, ..., Sink P). Each source has a finite capacity, each sink has a specified demand, and each mixer has limits on how much flow it can handle. Additionally, each sink imposes quality constraints (e.g., limiting the maximum allowable pollutant concentration). The objective is to determine the flow distribution that minimizes the total operating cost, which may include pumping costs, pipeline costs, and treatment costs if the water requires quality adjustment before reaching the sinks. Formally, let $f_{i,k}$ represent the flow rate from source i to mixer k , $g_{k,j}$ the flow from mixer k to sink j , and $z_{k,l}$ the flow between mixers k and l . Denote the cost coefficients by $c_{i,k}$, $c_{k,j}$, $c_{k,l}$, representing the cost of transporting water (and possibly performing intermediate treatment) along each respective pathway. The decision variables $f_{i,k}$, $g_{k,j}$, $z_{k,l}$ must be non-negative. Key parameters include the source capacities Q_i , sink demands D_j , maximum sink quality limits T_j , and mixer capacities. The total cost of operating the network is given by:

Equation 3-1 Source-Mixer-Sink Network Model for the Water/Wastewater Reuse Optimization Problem

* **Example Case:** Suppose a city has two water sources (with capacities $Q_1 = 50 \text{ m}^3/\text{h}$ and $Q_2 = 40 \text{ m}^3/\text{h}$), one mixer (capacity $80 \text{ m}^3/\text{h}$) for preliminary blending, and two industrial sinks (demands $D_1 = 30 \text{ m}^3/\text{h}$ and $D_2 = 50 \text{ m}^3/\text{h}$). Each sink also has a pollutant concentration limit (e.g., the maximum concentration at Sink 1 is $T_1 = 10 \text{ mg/L}$, and for Sink 2 it is $T_2 = 15 \text{ mg/L}$). By formulating the above constraints, the decision-maker can solve for $f_{1,k}$, $f_{2,k}$, $g_{k,1}$, $g_{k,2}$, and any potential $z_{k,l}$ flows if multiple mixers existed. The optimization might reveal that to keep the concentration at Sink 1 under 10 mg/L , more of the cleaner source water is routed to that sink, whereas Sink 2 can handle higher pollutant levels but requires more flow to meet its larger demand. Ultimately, the solution identifies the minimum transportation and treatment costs that satisfy capacity, demand, and quality requirements.

$$\text{Minimize } Z = \sum_{i=1}^M \sum_{k=1}^N c_{i,k} f_{i,k} + \sum_{k=1}^N \sum_{j=1}^P c_{k,j} g_{k,j} + \sum_{k=1}^N \sum_{l=1}^N c_{k,l} z_{k,l}.$$

Subject to this objective, we impose several constraints:

- **Flow Balance at Mixers:** The total inflow to each mixer equals its total outflow. For every mixer $k \in \{1, \dots, N\}$,

$$\sum_{i=1}^M f_{i,k} + \sum_{l=1}^N z_{l,k} = \sum_{j=1}^P g_{k,j} + \sum_{l=1}^N z_{k,l}.$$

- **Source Capacity:** The total flow from each source i cannot exceed its capacity Q_i :

$$\sum_{k=1}^N f_{i,k} \leq Q_i, \forall i \in \{1, \dots, M\}.$$

- **Sink Demand:** The total flow into each sink j must meet or exceed the demand D_j :

$$\sum_{k=1}^N g_{k,j} \geq D_j, \forall j \in \{1, \dots, P\}.$$

- **Quality Constraints at Sinks:** Let C_k represent the (possibly blended) pollutant concentration of water leaving mixer k . Then, for each sink j ,

$$\sum_{k=1}^N (g_{k,j} \cdot C_k) \leq T_j \cdot \sum_{k=1}^N g_{k,j}.$$

This enforces that the weighted-average pollutant concentration at sink j does not exceed T_j .

- **Mixer Flow Capacity:** Each mixer k has a maximum throughput limit (*Mixer Capacity*), which constrains the sum of flows entering it:

$$\sum_{i=1}^M f_{i,k} + \sum_{l=1}^N z_{l,k} = \text{Mixer Capacity}, \forall k \in \{1, \dots, N\}.$$

- **Non-negativity:** All flows must be non-negative,

$$f_{i,k}, g_{k,j}, z_{l,k} \geq 0.$$

3.5.2 Multi-stage Wastewater Treatment Plant Network Design Problem

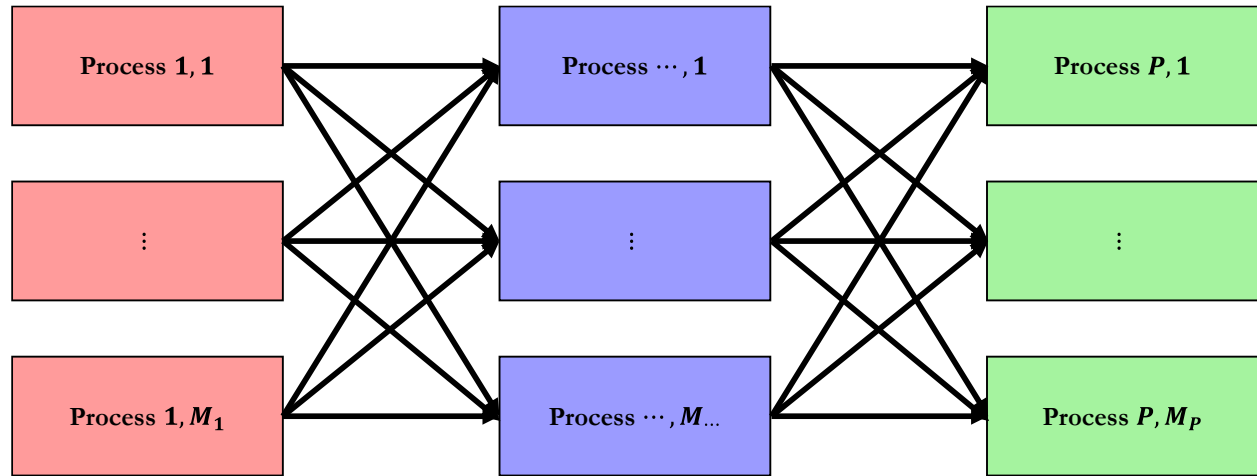


Figure 3-4 Multi-stage wastewater treatment plant network design problem superstructure concerning multiple interlinked process units.

The second, more holistic problem is on the design of a comprehensive “superstructure” for a wastewater treatment plant (WWTP), where each stage of treatment can be performed by one of several possible processes. The plant is divided into P sequential stages, and each stage p has M_p candidate processes (e.g., different types of clarifiers, biological reactors, or membrane units). Our decision is both which process to select at each stage and how much flow to pass through it, thereby creating an optimal chain of processes that achieves the desired effluent quality with minimal total cost. This problem has been widely studied in the literature in various applications (Galan & Grossmann, 1998; Galán & Grossmann, 1999a, 1999b; Tosarkani & Amin, 2020; L. Yang et al., 2014; Ye et al., 2019). A general source-sink presentation[†] of the problem is shown in **Figure 3-4** and its mathematical formulation is as follows:

[†] **Example Case:** Consider a WWTP that must remove both chemical oxygen demand (COD) and nitrogen from an influent stream of 1000 m³/day. Suppose it has three stages: Stage 1 can be either a primary clarifier or a coarse filtration unit; Stage 2 can be either a conventional activated sludge reactor or a moving bed bioreactor; Stage 3 can be either a settling tank or a membrane filtration unit. Each process has a different removal efficiency for COD and nitrogen, a specific capacity constraint, and a cost coefficient $Cost_{p,m}$. By constructing the binary selection variables $x_{p,m}$ and flow variables $f_{p,m}$, one can systematically evaluate all combinations (superstructure) to find the sequence and flow distribution that best meets the final effluent standards (e.g., COD < 30 mg/L, total nitrogen < 10 mg/L) at minimal total cost. The solution might reveal, for instance, that choosing a primary clarifier in Stage 1 and a membrane filtration unit in Stage 3 is worth a higher capital cost because it results in lower operational expenses, ultimately reducing the total cost across the plant’s lifetime.

In this superstructure model, we introduce a binary variable $x_{p,m}$ to indicate whether process m at stage p is chosen ($x_{p,m} \in \{0,1\}$) and a continuous variable $f_{p,m}$ for the flow through that process. Each process has an operating cost $Cost_{p,m}$, which can represent the sum of investment, operational, and maintenance expenses. The goal is to minimize:

Equation 3-2 Source- Sink Optimization Model for the Multi-stage Wastewater Treatment Plant Design Problem

$$\text{Minimize } Z = \sum_{p=1}^P \sum_{m=1}^{M_p} (Cost_{p,m} \cdot x_{p,m}).$$

Subject to:

- **Process Selection:** Exactly one process must be chosen at each stage:

$$\sum_{m=1}^{M_p} x_{p,m} = 1, \forall p \in \{1, \dots, P\}.$$

- **Flow Balance:** The total flow leaving stage $p - 1$ must equal the total flow entering stage p :

$$\sum_{m=1}^{M_{p-1}} f_{p-1,m} = \sum_{m=1}^{M_p} f_{p,m}, \forall p \in \{2, \dots, P\}.$$

- **Capacity Constraints:** The flow through process mm at stage p must not exceed the capacity Q_p of that stage, and it applies only if that process is selected:

$$f_{p,m} \leq Q_p \cdot x_{p,m}, \forall p, m$$

- **Pollutant Removal:** Let $c_{p,m}$ denote the effluent pollutant concentration after process m at stage p . If the influent concentration to stage p is $c_{p-1,m}$ (for some process mm in stage $p - 1$), then pollutant removal efficiency $C_{p,m}$ gives:

$$c_{p,m} = c_{p-1,m} (1 - C_{p,m}) \cdot x_{p,m}, \forall p \geq 2, m.$$

Here, $(1 - C_{p,m})$ captures the fraction of pollutant remaining after the treatment step.

- **Final Quality Requirement:** At the final stage P , the total pollutant load in the combined outflow must be below the permissible limit T_{max} . One common representation is:

$$\sum_{m=1}^{M_P} c_{P,m} \cdot f_{P,m} \leq T_{max}.$$

- **Non-negativity and Binary Variables:**

$$f_{p,m} \geq 0, c_{p,m} \geq 0, x_{p,m} \in \{0,1\}.$$

3.5.3 Nonlinear Process Optimization for a Single Biological Treatment Unit

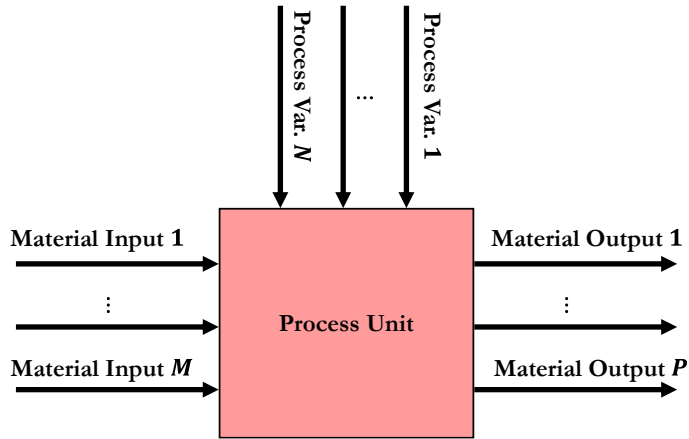


Figure 3-5 Process Optimization Superstructure for a Single Biological Treatment Unit.

The third problem focuses on a single biological wastewater treatment unit in which pollutant removal is strongly influenced by both operational variables and the influent pollutant composition. This is a widely researched topic in the field of biological wastewater treatment (Ahn et al., 2014; Asadi et al., 2017; Fang et al., 2011; Kim et al., 2009; Kusiak & Wei, 2012; Lim et al., 2011; Wu et al., 2016). The simplistic superstructure representation of the optimization problem is presented[‡] in **Figure 3-5**, while the general mathematical formulation of the optimization model is presented as follows:

Let there be M distinct pollutants (Pollutant 1,...,Pollutant M) entering the reactor and N process variables (Var. 1,...,Var. N) that can be adjusted, such as aeration rate, sludge age, or nutrient dosing. Denote each process variable as x_n , bounded by $[x_n^{min}, x_n^{max}]$. We assume the influent concentration of pollutant m is I_m , and the effluent concentration O_m depends on a removal efficiency function $R_m(x_1, x_2, \dots, x_N, I_1, \dots, I_M)$. Because

[‡] **Example Case:** Consider a single activated sludge reactor that must treat an influent containing two pollutants: organic matter (COD) and ammonia. Two adjustable parameters might be the dissolved oxygen level x_1 and the sludge retention time x_2 . The removal efficiencies for COD and ammonia, $R_{COD}(x_1, x_2)$ and $R_{NH_3}(x_1, x_2)$, are strongly interdependent. Increasing dissolved oxygen might improve ammonia removal but could raise operational costs due to higher aeration demand, while a longer sludge age generally improves both removals but leads to higher sludge handling costs. Formulating this as a constrained optimization problem with the objective of minimizing total operating cost (aeration plus sludge handling) allows the plant operator to find the optimal balance: for instance, setting the dissolved oxygen at 2 mg/L and sludge age at 10 days might meet all effluent limits at the lowest overall cost. Changes in influent composition (e.g., higher ammonia load) would shift the optimal operating point, showcasing the need for repeated or adaptive optimization.

biological systems involve complex, nonlinear interactions (e.g., microbial kinetics, competition for substrates), this removal efficiency can be a complicated function of both the process variables and the influent pollutant levels. The objective is to minimize the total operational cost, typically expressed as a function of the process variables multiplied by the wastewater flow rate Q . For instance, if C_n is the cost coefficient (in \$/unit) for variable x_n , the objective function can be written as:

Equation 3-3 *Process Optimization Model for a Single Biological Treatment Unit*

$$\text{Minimize } Z = Q \cdot \sum_{n=1}^N C_n x_n.$$

Subject to:

- **Effluent Quality Constraints:** Each pollutant's final concentration must meet the regulatory limit E_m :

$$O_m \leq E_m, \forall m \in \{1, \dots, M\}.$$

- **Effluent Concentration Relationship:** The effluent concentration is the influent concentration times the fraction that remains after treatment:

$$O_m = I_m (1 - R_m(x_1, x_2, \dots, x_N, I_1, \dots, I_M)), \forall m.$$

- **Process Variable Bounds:** Each x_n must lie within feasible operating ranges:

$$x_n^{\min} \leq x_n \leq x_n^{\max}, \forall n \in \{1, \dots, N\}.$$

- **Removal Efficiency Constraints:** Physically valid removal efficiencies lie between 0 and 1:

$$0 \leq R_m(x_1, \dots, x_N, I_1, \dots, I_M) \leq 1, \forall m.$$

Since R_m often includes terms reflecting microbial growth rates, Monod kinetics, or other empirical relationships, the problem is inherently nonlinear and potentially nonconvex. Solving such a model typically involves employing advanced nonlinear optimization algorithms that can handle the potential complexity of the solution space.

3.6 ML Models in Optimizing Biological Wastewater Treatment

Over the past two decades, the application of advanced machine learning (ML) techniques to optimize biological wastewater treatment processes has gained considerable momentum (Croll et al., 2023; Nazlabadi et al., 2021). Traditional process optimization approaches often relied on mechanistic modeling and linear or polynomial-based regression methods (Nazlabadi et al., 2021). However, the increasingly complex dynamics of biological treatment systems—characterized by multiple interdependent

compounds and nonlinear interactions—have prompted researchers to explore more flexible and data-driven ML models (Asadi et al., 2017). Notably, support vector machines (SVM), XGBoost, LightGBM, random forests, multi-level perceptrons (MLP), artificial neural networks (ANN), multi-adaptive regression splines (MARS), and k-nearest neighbors (k-NN) have emerged as compelling alternatives due to their ability to handle large datasets and capture highly nonlinear relationships among different process variables (Asadi et al., 2017; Fang et al., 2011; Kusiak & Wei, 2012; Lim et al., 2011; Wu et al., 2016).

One principal advantage of using ML algorithms in biological wastewater treatment is their capacity to model complex processes without requiring a comprehensive mechanistic understanding of the underlying biochemical reactions. Several studies illustrate this benefit. For example, Asadi et al. (2017) harnessed real-world process data to create a predictive model of the aeration process, a critical cost and energy driver in most wastewater treatment plants. Similarly, Kusiak and Wei (2012) employed ML-based approaches to optimize operational parameters in activated sludge processes, ultimately demonstrating a potential reduction in both energy consumption and chemical usage. These successes stem from the fact that ML models, through their sophisticated learning algorithms, can identify subtle, nonlinear, and potentially high-order interactions between factors such as oxygen transfer rates, influent loading, microbial community behavior, and environmental conditions. A crucial step in developing robust ML models for wastewater treatment optimization involves collecting sufficient and reliable training data. While some research groups rely on actual plant or laboratory-scale operational data, others have leveraged simulation packages like Biowin, GPS-X, AQUASIM, SUMO, and SIMBA (Ahn et al., 2014; Fang et al., 2011; Wu et al., 2016). These software platforms offer pre-calibrated mechanistic models of biological treatment processes, enabling researchers to generate large volumes of data under various hypothetical operating conditions (Bressani-Ribeiro et al., 2021; Caro et al., 2024; Díez-Montero et al., 2019; Hellal & Abou-Elela, 2021; Jasim, 2020; Kasi et al., 2011; Liang et al., 2021). This simulation-based approach can save time and resources compared to purely experimental data collection, though it relies on the assumption that the simulation models are accurately calibrated and reflective of real-world scenarios (Croll et al., 2023). The comprehensive datasets derived from these simulations provide a broad parameter space that ML models can learn from, enabling them to make more reliable predictions when faced with new or unseen conditions.

Once adequately trained, ML models are typically embedded within broader optimization frameworks that seek to identify the best set of decision variables, such as aeration rates, recycle ratios, nutrient dosing schedules, or sludge retention times (Ahn et al., 2014; Fang et al., 2011; Wu et al., 2016). Due to the highly nonlinear nature of many ML algorithms, conventional gradient-based optimization methods can be less effective or even inapplicable, as the solution space may be riddled with local minima or saddle points (Jain

& Kar, 2017). Consequently, researchers often resort to heuristic or meta-heuristic optimization strategies, including genetic algorithms, particle swarm optimization, and other evolutionary methods (Ahn et al., 2014; Fang et al., 2011; Wu et al., 2016). These approaches can navigate complex and discontinuous objective functions and constraints, offering “good-enough” solutions in a reasonable timeframe. However, they do not guarantee global optimality and may exhibit issues with convergence, especially as the search space becomes higher-dimensional or when the underlying ML model is extremely complex (Jain & Kar, 2017).

Despite its popularity, especially in earlier studies (Fang et al., 2011; Kim et al., 2009; Lim et al., 2011; Wu et al., 2016), multivariable response surface methodology (MRS) is scrutinized for inadequately addressing the interdependencies between different compounds or performance indicators. Many MRS models (Fang et al., 2011; Kim et al., 2009; Lim et al., 2011; Wu et al., 2016) assume that each compound or treatment performance metric can be optimized independently, effectively ignoring the coupled biochemical reactions and microbial community interactions inherent in biological systems. Such oversimplification can lead to suboptimal or misleading conclusions, as changing one parameter (e.g., carbon source addition) to optimize the removal of a particular compound (e.g., COD) may adversely affect the removal of another compound (e.g., nitrogen). A related methodological pitfall appears in studies that attempt to capture the interdependencies among wastewater constituents by including compound concentrations (the target variables) directly in the training dataset as inputs (Asadi et al., 2017; Kusiak & Wei, 2012). This practice can lead to “data spillage,” where the predictive model inadvertently learns from outcomes that should be unknown during the training phase. This can artificially inflate model accuracy and reduce generalizability to real operational settings. When used in optimization, these flawed models risk recommending untrustworthy operating conditions that fail to meet performance requirements when deployed in practice. On the other hand, while advanced ML models excel at capturing nonlinearities, they generally lack the mechanistic interpretability of simpler linear or phenomenological models. Their “black-box” nature makes it challenging to derive physical or biochemical insights, limiting the scope for process engineers to leverage domain knowledge or identify fundamental process improvements (Croll et al., 2023). Because biological wastewater treatment is inherently governed by microbial ecology, biochemical kinetics, and mass transfer phenomena, the inability to interpret model coefficients or internal architecture often complicates efforts to streamline or re-engineer the process based on fundamental scientific principles. It should also be noted the no application of any of the mentioned ML methods have been proven effective for modeling Advective-Diffusive attached-growth systems like bio-towers, which is a major gap in the literature.

Another challenge arises when extending optimization efforts beyond a single treatment unit to multiple interlinked units, or a superstructure involving multiple sources, mixers,

and sinks such as those in **Equation 3-1**. In such scenarios, the number of decision variables and constraints multiplies quickly, making the overall optimization problem computationally expensive and prone to convergence issues—especially if a heuristic optimizer is used in conjunction with a complex ML model. Ensuring that these large-scale optimization problems remain tractable without compromising solution accuracy is a prominent concern for real-world implementations. In light of these challenges, there is a growing call for developing models that offer the best of both worlds: (i) the ability to capture interdependencies among wastewater constituents, (ii) the avoidance of data spillage, (iii) mechanistic interpretability, and (iv) predictive power that rivals or surpasses that of nonlinear ML methods. A linear model that is carefully designed to reflect the interactions of different compounds would address many of the criticisms leveled against purely data-driven ML approaches (Nazlabadi et al., 2021). By incorporating terms or factors that account for relevant biochemical relationships, such a model could maintain interpretability—offering straightforward parameter coefficients tied to physical phenomena—while still accurately predicting system behavior. This is especially critical in large-scale optimization tasks, such as superstructure design, water reuse networks (**Equation 3-1**), and multi-stage wastewater treatment plant designs (**Equation 3-2**), where linear (or linearized) models are often necessary for computational tractability (Nazlabadi et al., 2021). Furthermore, a linear model that respects the avoidance of data spillage would ensure more rigorous calibration and validation processes. By carefully segregating inputs and outputs—such that the target variables do not inadvertently enter as model inputs—researchers could preserve the model’s integrity and bolster confidence in its extrapolative capabilities. This methodological rigor would also facilitate clearer integration of domain knowledge about microbial kinetics or stoichiometric relationships, potentially bridging the gap between data-driven and mechanistic modeling approaches.

4 Statement of the Problem

Over the past two decades, various machine learning (ML) techniques have demonstrated promise in modeling and optimizing biological wastewater treatment processes. These approaches can capture nonlinear relationships among operational parameters, biochemical reactions, and treatment performance indicators. However, three main challenges persist:

Data Spillage and Oversimplification

Many existing studies include performance indicators (e.g., effluent concentrations) as direct inputs to ML models, leading to “data spillage.” This undermines model generalizability and inflates predictive accuracy in ways that do not hold in real operational settings. Additionally, linear or polynomial models often fail to capture compound interdependencies, ignoring the coupled biochemical reactions and microbial community interactions critical to process performance.

Lack of Mechanistic Interpretability

Advanced ML methods (e.g., random forests, neural networks) deliver high predictive power but function as “black boxes,” making it difficult to extract mechanistic insights. This lack of transparency limits the ability of process engineers to incorporate domain knowledge, improve fundamental processes, or troubleshoot effectively when real-world performance deviates from model predictions.

Scalability and Model Gaps

When extending optimization to large-scale treatment superstructures with multiple units, sources, mixers, and sinks, the dimensionality of the decision space escalates, causing heuristic optimizers and complex ML models to struggle with convergence and tractability. Moreover, no ML method has been validated for modeling advective-diffusive attached-growth systems (e.g., bio-towers), leaving an important class of wastewater processes insufficiently studied.

These gaps underscore the need for a modeling framework that:

- Reflects the interconnected nature of multiple wastewater constituents without causing data spillage,
- Maintains interpretability by integrating fundamental biochemical relationships, and
- Remains computationally tractable for large-scale or superstructure optimization tasks.

The research questions that this dissertation will answer are the following:

- How can ML-based models be designed to capture the interdependencies among multiple wastewater constituents without incurring data spillage, thereby improving predictive accuracy and generalizability?
- What modeling strategies or frameworks can balance the “black-box” nature of advanced ML methods with the mechanistic interpretability needed for practical process design and troubleshooting?
- How can these improved modeling approaches be scaled up to large, multi-stage or superstructure configurations—including advective-diffusive attached-growth systems—while ensuring computational feasibility and robust convergence in optimization?

The primary objective of this dissertation is to develop and validate a modeling and optimization framework that bridges the gap between purely data-driven ML methods and mechanistic models for biological wastewater treatment. Specifically, the proposed approach will:

- Eliminate data spillage by rigorously separating input variables from performance metrics in model training and validation, ensuring more trustworthy predictions.
- Enhance mechanistic interpretability by incorporating biochemical and microbial process fundamentals into the modeling architecture, providing process engineers with actionable insights.
- Demonstrate scalability through the application of the framework to multi-stage systems and superstructures, including advective-diffusive attached-growth units like bio-towers, thereby confirming both computational tractability and practical relevance.

By achieving these goals, the study aims to lay a foundation for next-generation modeling tools that offer robust, interpretable, and scalable solutions for optimizing biological wastewater treatment processes.

5 Significance of the Study

This study holds significant value on both theoretical and practical fronts by addressing the critical limitations in existing modeling and optimization approaches for biological wastewater treatment. First, it responds to the persistent issue of “data spillage,” whereby models are inadvertently trained using the same performance indicators they are tasked to predict. By proposing a rigorous strategy to separate input parameters from output variables during model development, this research establishes a methodological safeguard that greatly enhances the generalizability and reliability of predictions. The avoidance of data spillage is particularly important in operational settings where overly optimistic model performance often fails to translate into real-world success. Through its emphasis on methodological integrity, the study’s contributions will help process engineers make more dependable decisions about treatment system design and daily operations.

Second, this work advances the field by reconciling advanced machine learning (ML) techniques with mechanistic interpretability. While black-box ML algorithms such as neural networks and ensemble methods can capture highly nonlinear relationships, they often obscure the biochemical and microbial interactions that govern treatment performance. By integrating fundamental biochemical and microbial principles into the modeling architecture, this dissertation demonstrates how interpretability and domain knowledge can be preserved without sacrificing predictive accuracy. This synergy is crucial for engineers and researchers who not only need accurate predictions but also require clear pathways to troubleshoot issues and refine plant operations based on physical or biological insights. The enhanced interpretability opens the door to more transparent design revisions, better-informed regulatory compliance, and more targeted process improvements.

A third key contribution lies in the framework’s demonstrated scalability to large-scale or superstructure configurations—a critical aspect for municipalities, industrial operators, and research institutions seeking to optimize complex treatment networks. Existing methods often falter when the dimensionality of a multi-unit system or superstructure becomes too large, leading to insurmountable computational demands and convergence problems. By employing linear or carefully linearized models that still respect the underlying biochemical relationships, the proposed approach maintains computational tractability, making large-scale optimization feasible. Notably, this framework also covers advective-diffusive attached-growth systems like bio-towers, which have not yet been modeled successfully by current ML techniques. Filling this gap extends the applicability of next-generation modeling tools to a wider range of treatment processes, ultimately helping operators achieve more holistic and efficient treatment designs.

Finally, by blending data-driven and mechanistic approaches, the dissertation paves the way for future research in hybrid modeling and advanced control of wastewater processes. Its methodological rigor, especially in terms of careful calibration, validation, and interpretability, not only establishes a template for improved modeling practices but also lays a foundation for the real-time optimization of treatment plants. The potential impacts include reducing operational costs, minimizing energy consumption, ensuring reliable effluent quality, and accommodating evolving regulatory standards. Collectively, these contributions position this study at the forefront of efforts to modernize and enhance wastewater treatment technologies, ensuring more sustainable and effective solutions for safeguarding public health and the environment.

6 Scope and Limitations of the Study

This dissertation is dedicated to the development and validation of a hybrid modeling and optimization framework specifically designed for biological wastewater treatment systems. The research focuses on integrating fundamental biochemical principles with advanced machine learning (ML) techniques to achieve a balance between high predictive accuracy and mechanistic interpretability. In particular, the framework is demonstrated using two key treatment processes: (1) the activated sludge process, representing suspended-growth systems, and (2) bio-towers, representing attached-growth systems. These two processes collectively capture a broad spectrum of biological treatment scenarios—ranging from conventional aeration basins to biofilm-based reactors—and therefore serve as representative testbeds for evaluating the robustness, scalability, and interpretability of the proposed modeling approach. Although limiting case studies to these two processes excludes other potential treatment technologies (e.g., membrane bioreactors, sequencing batch reactors), they were selected because of their widespread application, data availability in published literature, and capacity to illustrate both suspended- and attached-growth phenomena.

A defining characteristic of this study is the reliance on experimental data sourced exclusively from existing literature, rather than generating new experimental data. This approach makes it possible to incorporate a diverse range of documented operational scenarios and performance metrics from multiple studies, thereby enhancing the framework's potential applicability across various plant configurations and treatment objectives. However, this choice introduces certain limitations. First, the quality and consistency of the secondary data can vary considerably, influencing both the calibration and validation phases of the modeling process. Ensuring compatibility and coherency among datasets derived from disparate sources is an inherent challenge that must be carefully managed to maintain methodological rigor. Second, the decision to focus on the activated sludge process and bio-towers means that some advanced or specialized wastewater technologies (e.g., membrane reactors or emerging hybrid systems) will not be covered in depth. While this targeted approach is sufficient for illustrating the feasibility and benefits of the proposed framework, it may limit generalizability to other treatment processes that involve substantially different biochemical or operational complexities.

Additionally, while the study leverages a wide range of operational conditions, it cannot represent every conceivable variation found in real-world wastewater treatment practice. This selective coverage is necessary to maintain a focused scope and to demonstrate depth of analysis in the target domains; nevertheless, it also highlights the need for future research to extend or adapt the framework to other treatment technologies. The computational feasibility of the proposed optimization algorithms presents another

constraint. Although the framework is designed to handle large-scale systems—especially superstructures that involve multiple sources, sinks, and interconnected units—practical deployment for real-time optimization or extremely large configurations may require advanced computing resources that exceed the scope of this work.

Another important limitation relates to the balance between interpretability and predictive power. While the hybrid approach strives to merge mechanistic insights with data-driven modeling to facilitate transparency, certain black-box ML algorithms might deliver slightly higher predictive accuracy for highly specialized tasks at the expense of clarity about underlying biochemical processes. This trade-off is purposeful, ensuring that the insights provided by the model can be acted upon by process engineers who rely on mechanistic understanding for effective troubleshooting and process optimization. Furthermore, although the study includes bio-towers as a representative of advective-diffusive attached-growth systems, the inherent variability and unique conditions of individual treatment plants imply that further calibration and customization may be required to realize optimal performance in specific operational contexts.

Despite these constraints, the scope of this dissertation is well-justified by the pressing need to develop a robust, scalable, and interpretable modeling framework that addresses critical gaps in current wastewater treatment research. By focusing on both suspended-growth (activated sludge) and attached-growth (bio-towers) processes, and by employing a breadth of existing experimental data, the study maximizes its relevance to real-world situations while also laying a solid foundation for future extensions.

7 Model Formulation

7.1 Coupled Linear Equations Framework for Optimization (CLEFO)

The biochemical processes involved in biological wastewater treatment are inherently complex, characterized by multiple interdependent constituents, including various compounds and biomass concentrations. Here, we propose a coupled linear equations framework for optimization tailored to encapsulate the intricate interdependencies and interaction effects inherent in biological wastewater treatment systems. The primary objective of this model is to provide a tractable yet comprehensive framework that captures the dynamic relationships among active wastewater constituents, specifically compound and biomass concentrations, while accounting for the influence of key operational parameters such as flow rate, media porosity, and influent compound concentrations without loss of tractability in a global optimization routine.

7.2 Variables

The model delineates independent variables, encompassing factors like flow rate, media porosity, and process variables, denoted as $\mathbf{X}_p$, and various influent compound concentrations, denoted as $\mathbf{X}_c$. To concatenate $\mathbf{X}_p$ (process variables) and $\mathbf{X}_c$ (influent compound concentrations), we can define a single vector $\mathbf{X} = \begin{bmatrix} \mathbf{X}_p \\ \mathbf{X}_c \end{bmatrix} = [X_1, X_2, \dots, X_o, X_{o+1}, X_{o+2}, \dots, X_m]^T$ by stacking $\mathbf{X}_p$ and $\mathbf{X}_c$ vertically. This ensures that all independent variables are represented in a unified manner. These variables serve as the primary drivers influencing the system's behavior. Concurrently, the model identifies dependent variables ($\mathbf{Y}$), which include post-treatment compound concentrations ($Y_1, Y_2, \dots, Y_p$) and biomass concentrations ($Y_{p+1}, Y_{p+2}, \dots, Y_n$). Recognizing the inherent interdependencies among these dependent variables—where the concentration of one compound or biomass species may influence others—the model incorporates a coupling coefficients matrix ($\mathbf{\Gamma}$) to represent these mutual interactions. For example, in a biological wastewater treatment system, the concentration of dissolved oxygen (Y_1) plays a critical role in the activity and growth of aerobic biomass species, such as nitrifying bacteria (Y_2). An increase in Y_1 can enhance the metabolic activity of nitrifying bacteria, leading to higher rates of ammonium (Y_3) oxidation to nitrite and nitrate. Conversely, as the nitrification process progresses and oxygen is consumed, Y_1 may decrease, potentially limiting the activity of the aerobic biomass. This dynamic interplay between dissolved oxygen and biomass illustrates how changes in one dependent variable can cascade through the system, influencing others. The coupling coefficients matrix ($\mathbf{\Gamma}$) captures these interdependencies quantitatively, enabling the model to account for such feedback loops and their effects on the overall biochemical processes.

On the other hand, to capture the interaction effects among independent variables, the model introduces pairwise interaction terms ($\mathbf{Z}$). These are constructed by taking the product of every unique pair of independent variables, such as the interaction between flow rate and media porosity ($Z_1 = X_1 \times X_2$), thereby encapsulating synergistic or antagonistic effects that individual variables alone may not reveal. For instance, in a biological wastewater treatment system, the interaction between flow rate (X_1) and media porosity (X_2) can significantly influence the efficiency of the treatment process. A high flow rate might lead to insufficient contact time between the influent wastewater and the treatment media, potentially reducing the removal efficiency of pollutants. However, if the media porosity is also high, it could compensate for the reduced contact time by allowing greater penetration and distribution of wastewater throughout the treatment bed. Conversely, a combination of low flow rate and low porosity might create stagnant zones, leading to uneven treatment and localized inefficiencies. By including the interaction term ($Z_1 = X_1 \times X_2$), the model captures these synergistic or antagonistic effects, reflecting how the interplay between flow rate and media porosity jointly impacts the treatment performance in ways that considering these variables individually might miss.

Additionally, the model accounts for interaction effects between independent and dependent variables through another set of interaction terms ($\mathbf{W}$), defined as the product of each independent variable with each dependent variable (e.g., $W_1 = X_1 \times Y_1$). This allows the influence of an independent variable on a dependent variable to be modulated by the current state of that dependent variable, effectively capturing feedback mechanisms within the system. For example, consider the interaction between flow rate (X_1) and the concentration of dissolved oxygen (Y_1) in a biological wastewater treatment system. While a higher flow rate can increase the mixing of oxygen in the system, thereby potentially raising dissolved oxygen levels, the actual impact depends on the existing concentration of Y_1 . If Y_1 is already at a sufficient level, the additional mixing from increased X_1 may have a negligible effect. Conversely, if Y_1 is low, the same increase in X_1 could have a significant positive impact by enhancing oxygen transfer. By including the interaction term $W_1 = X_1 \times Y_1$, the model captures this feedback mechanism, where the effect of flow rate on dissolved oxygen is modulated by the current state of Y_1 .

7.3 Coupled Linear Equations Model

The mathematical formulation of the baseline model is expressed as:

Equation 7-1 Coupled linear equations model

$$\mathbf{Y} = \mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{\Theta}\mathbf{Z} + \mathbf{\Gamma}\mathbf{Y} + \text{diag}(\mathbf{\Lambda}\mathbf{X})\mathbf{Y}.$$

In this equation, the intercept vector $\mathbf{Y} = [\epsilon_1, \epsilon_2, \dots, \epsilon_n]^T$ is an $n \times 1$ vector where each element ϵ_i represents the intercept or residual for the i -th dependent variable Y_i . It provides the baseline level of each dependent variable when all independent variables and interaction terms are zero. On the other hand, the linear coefficients matrix $\mathbf{B}$ is an

$n \times m$ matrix where each element $\beta_{i,j}$ quantifies the direct influence of the j -th independent variable X_j on the i -th dependent variable Y_i .

$$\mathbf{B} = \begin{bmatrix} \beta_{1,1} & \beta_{1,2} & \cdots & \beta_{1,m} \\ \beta_{2,1} & \beta_{2,2} & \cdots & \beta_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{n,1} & \beta_{n,2} & \cdots & \beta_{n,m} \end{bmatrix}.$$

This matrix captures the linear relationships between each independent variable and each dependent variable, allowing the model to account for how changes in operational parameters directly affect treatment outcomes. We also introduce the interaction coefficients matrix for $\mathbf{Z}$ denoted by $\mathbf{\Theta}$, which is an $n \times q$ matrix, where $q = \binom{m}{2}$ represents the number of unique pairwise interaction terms among independent variables. Each element $\theta_{i,k}$ quantifies the effect of the k -th interaction term Z_k on the i -th dependent variable Y_i .

$$\mathbf{\Theta} = \begin{bmatrix} \theta_{1,1} & \theta_{1,2} & \cdots & \theta_{1,q} \\ \theta_{2,1} & \theta_{2,2} & \cdots & \theta_{2,q} \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{n,1} & \theta_{n,2} & \cdots & \theta_{n,q} \end{bmatrix}.$$

By incorporating interaction terms, this matrix allows the model to capture how combinations of independent variables jointly influence the dependent variables, reflecting synergistic or antagonistic effects that are not apparent when considering variables in isolation. On the other hand, we have the coupling coefficients matrix among $\mathbf{Y}$ denoted by $\mathbf{\Gamma}$, which is an $n \times n$ matrix where each element $\gamma_{i,j}$ represents the influence of the j -th dependent variable Y_j on the i -th dependent variable Y_i .

$$\mathbf{\Gamma} = \begin{bmatrix} \gamma_{1,1} & \gamma_{1,2} & \cdots & \gamma_{1,n} \\ \gamma_{2,1} & \gamma_{2,2} & \cdots & \gamma_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{n,1} & \gamma_{n,2} & \cdots & \gamma_{n,n} \end{bmatrix}.$$

This coupling matrix captures the interdependencies among dependent variables, allowing the concentration of one compound or biomass species to affect others. Such interactions are commonplace in biochemical processes where, for example, the growth of one biomass species may inhibit or promote the growth of another. Lastly, we have the interaction coefficients matrix between $\mathbf{X}$ and $\mathbf{Y}$ denoted by $\mathbf{\Lambda}$ is an $n \times m$ matrix where each element $\lambda_{i,j}$ quantifies how the j -th independent variable X_j interacts with the i -th dependent variable Y_i .

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,n} \\ \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{n,1} & \lambda_{n,2} & \cdots & \lambda_{n,n} \end{bmatrix}.$$

This matrix enables the model to account for how the effect of an independent variable on a dependent variable may vary depending on the current state of that dependent variable. For instance, the impact of flow rate on biomass concentration might differ based on the existing biomass levels, reflecting feedback mechanisms within the treatment system. On the other hand, the diagonal interaction matrix $\text{diag}(\Lambda\mathbf{X})$ is an $n \times n$ diagonal matrix where each diagonal element $(\Lambda\mathbf{X})_{i,i} = \sum_{j=1}^m \lambda_{i,j} X_j$ and thus:

$$\text{diag}(\Lambda\mathbf{X}) = \begin{bmatrix} \sum_{j=1}^m \lambda_{1,j} X_j & 0 & \cdots & 0 \\ 0 & \sum_{j=1}^m \lambda_{2,j} X_j & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sum_{j=1}^m \lambda_{n,j} X_j \end{bmatrix}.$$

By scaling each Y_i with $\sum_{j=1}^m \lambda_{i,j} X_j$, the model incorporates how the influence of each independent variable on a dependent variable is modulated by both the variable itself and the current state of the dependent variable. This design maintains model linearity while allowing for dynamic interaction effects.

$$(\text{diag}(\Lambda\mathbf{X})\mathbf{Y})_i = \sum_{j=1}^m \lambda_{i,j} X_j Y_i = Y_i \sum_{j=1}^m \lambda_{i,j} X_j.$$

The complete model equation can be rearranged to isolate the dependent variables $\mathbf{Y}$ as follows:

Equation 7-2 *Solution to the coupled linear equations model by matrix inversion*

$$\begin{aligned} (\mathbf{I} - \mathbf{\Gamma} - \text{diag}(\Lambda\mathbf{X}))\mathbf{Y} &= \mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{\Theta}\mathbf{Z} \\ \mathbf{Y} &= (\mathbf{I} - \mathbf{\Gamma} - \text{diag}(\Lambda\mathbf{X}))^{-1}(\mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{\Theta}\mathbf{Z}). \end{aligned}$$

In this equation, $\mathbf{I}$ is the identity matrix.

7.4 Single-Unit Optimization Framework

We shall use **Equation 7-1** to solve the single-unit design problem defined in **Equation 3-3** by developing a single-unit CLEFO method. The objective function of the optimization routine can be formulated as a multi-objective function representing the minimization of pollutant concentrations and operational costs using $[c_{1,1}, \dots, c_{1,n}]^T$ and $[c_{2,1}, \dots, c_{2,m}]^T$ as their respective weight vectors. We also define the dependent variable set $\mathcal{Y} = \{1, 2, \dots, n\}$ and independent variable set $\mathcal{X} = \{1, 2, \dots, m\}$. Given these, we can express the explicit nonlinear optimization model using the summation form of **Equation 7-1** as follows:

Equation 7-3 Single-unit nonlinear optimization framework

$$\begin{cases} \min \sum_{i \in \mathcal{Y}} c_{1,i} Y_i + \sum_{j \in \mathcal{X}} c_{2,j} X_j \\ \text{s. t.} \begin{cases} Y_i + \sum_{j \in \mathcal{Y}} \gamma_{i,j} Y_j + \sum_{j \in \mathcal{X}} \lambda_{i,j} X_j Y_i = \epsilon_i + \sum_{j \in \mathcal{X}} \beta_{i,j} X_j + \sum_{l \in \mathcal{X}} \sum_{k \in \mathcal{X}} \theta_{i,l,k} X_l X_k, & \forall i \in \mathcal{Y}. \\ X_j^{\min} \leq X_j \leq X_j^{\max}, & Y_i^{\min} \leq Y_i \leq Y_i^{\max}, & \forall i \in \mathcal{Y}, \quad \forall j \in \mathcal{X} \end{cases} \end{cases}$$

The bilinear terms of the optimization problem may be linearized using a plethora of approaches. However, the improvement of such linearization techniques may not be necessary given that the decision variables and constraints of the optimization problem scale only with the wastewater compounds and process variables considered. It is, however, worthwhile to investigate the gains obtained from a linear model. To this end, we first attempt to relax the bilinear terms $X_j Y_i$ and $X_l X_k$. Linearization of such multiplications of continuous variables can be complex and computationally demanding. We employ a relaxation approach for bounded variables by transforming the product of two continuous variables into a separable form (Bisschop, 2018). We employ this approach to relax the terms. Consider the product of two continuous variables, $x_1 \cdot x_2$. To obtain a separable form, we define two new continuous variables y_1 and y_2 as follows:

$$\begin{aligned} y_1 &= 0.5(x_1 + x_2), \\ y_2 &= 0.5(x_1 - x_2). \end{aligned}$$

By construction, the product $x_1 \cdot x_2$ can now be represented as a difference of squares:

$$y_1^2 - y_2^2 \equiv x_1 \cdot x_2.$$

Following the same principle, we apply this relaxation technique to the bilinear terms $X_j Y_i$ and $X_l X_k$ present in our optimization model. For each bilinear term $X_j Y_i$, we introduce two auxiliary variables:

$$\begin{aligned} u_{i,j,1} &= 0.5(X_j + Y_i), \\ u_{i,j,2} &= 0.5(X_j - Y_i). \end{aligned}$$

Using these definitions, the product $X_j Y_i$ is equivalently written as:

$$X_j Y_i = u_{i,j,1}^2 - u_{i,j,2}^2.$$

If $X_j^{\min} \leq X_j \leq X_j^{\max}$ and $Y_i^{\min} \leq Y_i \leq Y_i^{\max}$, then the bounds of $u_{i,j,1}$ and $u_{i,j,2}$ are:

$$\begin{aligned} 0.5(X_j^{\min} + Y_i^{\min}) &\leq u_{i,j,1} \leq 0.5(X_j^{\max} + Y_i^{\max}), \\ 0.5(X_j^{\min} - Y_i^{\max}) &\leq u_{i,j,2} \leq 0.5(X_j^{\max} - Y_i^{\min}). \end{aligned}$$

This transformation replaces the original bilinear term with a separable expression involving the squares of new variables. Similarly, for each term $X_l X_k$, we define:

$$v_{l,k,1} = 0.5(X_l + X_k),$$

$$v_{l,k,2} = 0.5(X_l - X_k).$$

The product $X_l X_k$ can then be expressed as:

$$X_l X_k = v_{l,k,1}^2 - v_{l,k,2}^2.$$

The bounds of $v_{l,k,1}$ and $v_{l,k,2}$ are:

$$0.5(X_l^{min} + X_k^{min}) \leq v_{l,k,1} \leq 0.5(X_l^{max} + X_k^{max}),$$

$$0.5(X_l^{min} - X_k^{max}) \leq v_{l,k,2} \leq 0.5(X_l^{max} - X_k^{min}).$$

Incorporating these relaxations into the optimization model, the system dynamics constraints become:

$$Y_i + \sum_{j \in \mathcal{Y}} \gamma_{i,j} Y_j + \sum_{j \in \mathcal{X}} \lambda_{i,j} (u_{i,j,1}^2 - u_{i,j,2}^2) = \epsilon_i + \sum_{j \in \mathcal{X}} \beta_{i,j} X_j + \sum_{l \in \mathcal{X}} \sum_{k \in \mathcal{X}} \theta_{i,l,k} (v_{l,k,1}^2 - v_{l,k,2}^2), \quad \forall i \in \mathcal{Y}.$$

To arrive at a fully linear formulation, we now apply a piecewise linear (multiple choice) approximation for each squared auxiliary variable ($u_{i,j,1}^2, u_{i,j,2}^2, v_{l,k,1}^2, v_{l,k,2}^2$). Consider any one of the auxiliary variables, generically denoted as w , whose square we need to approximate (i.e., $w \in \{u_{i,j,1}, u_{i,j,2}, v_{l,k,1}, v_{l,k,2}\}$). We can use the known bound of w , i.e., $w^{min} \leq w \leq w^{max}$, to partition the interval $[w^{min}, w^{max}]$ into K subintervals $\{b_0, b_1, \dots, b_{K-1}, b_K\}$ using exponential breakpoints so that each subinterval becomes progressively wider. Suppose $\mathcal{Q} = \{1, 2, \dots, K\}$. The breakpoints are generated using:

$$b_q = w^{min} + \left(\frac{q}{K-1} \right)^2 (w^{max} - w^{min}), \quad \forall q \in \mathcal{Q}.$$

The spacing between consecutive breakpoints increases exponentially, with smaller intervals near w^{min} and wider intervals near w^{max} . This method ensures $b_0 < b_1 < b_2 < \dots < b_{K-1} < b_K$. These breakpoints provide better resolution than breakpoints of equal interval in regions where smaller values dominate and greater flexibility in capturing behavior across a wider range of the variable. This is demonstrated in **Figure 7-1**.

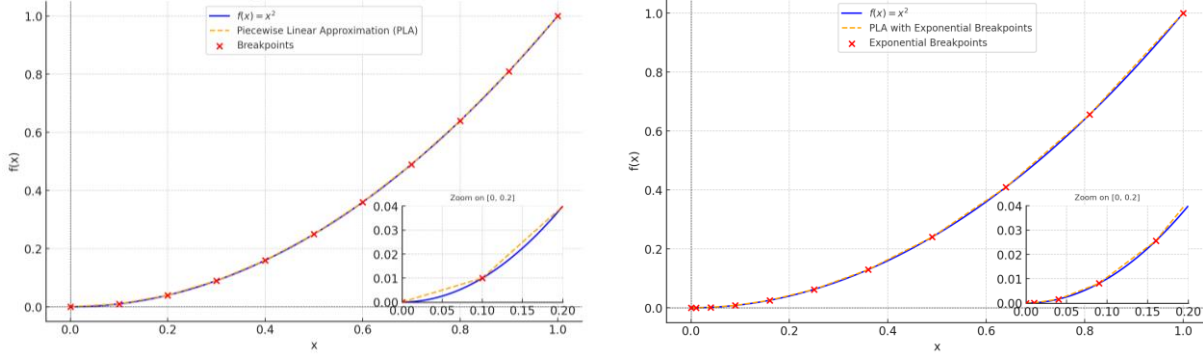


Figure 7-1 Comparison of linear (equal interval) and exponential breakpoints in piecewise linear approximation of quadratic variables (x^2)

To linearize w^2 over each subinterval $[b_{q-1}, b_q]$, we introduce a set of binary variables $\alpha_q \in \{0,1\}$ for all $q \in \mathcal{Q}$ and a continuous variable T that will approximate w^2 . We then impose two main sets of constraints:

$$\sum_{q \in \mathcal{Q}} \alpha_q = 1,$$

ensuring that w is assigned to exactly one of the K subintervals.

$$b_{q-1} \alpha_q \leq w \leq b_q \alpha_q, \quad \forall q \in \mathcal{Q}.$$

These inequalities restrict w to lie within $[b_{q-1}, b_q]$ only if $\alpha_q = 1$. For any other interval, those bounds become inactive. To complete the linearization, we must approximate the function $f(w) = w^2$ using a linear segment within each subinterval. One typical approach is to connect the points (b_{q-1}, b_{q-1}^2) and (b_q, b_q^2) with a chord. Denoting:

$$f(b_{q-1}) = b_{q-1}^2, \quad f(b_q) = b_q^2, \quad \Delta b_q = b_q - b_{q-1}, \quad \Delta f_q = b_q^2 - b_{q-1}^2,$$

the slope of this linear chord is

$$s_q = \frac{\Delta f_q}{\Delta b_q} = \frac{b_q^2 - b_{q-1}^2}{b_q - b_{q-1}} = b_q + b_{q-1}.$$

Hence, within the subinterval $[b_{q-1}, b_q]$, the linear approximation of w^2 can be written as:

$$f_q(w) = b_{q-1}^2 + s_q (w - b_{q-1}),$$

which we enforce via an additional continuous variable T in the following manner:

$$\begin{aligned} T - [b_{q-1}^2 + s_q (w - b_{q-1})] &\leq M (1 - \alpha_q), \\ [b_{q-1}^2 + s_q (w - b_{q-1})] - T &\leq M (1 - \alpha_q), \end{aligned}$$

where M is a sufficiently large constant (big- M parameter) for the q -th subinterval. These constraints ensure that if $\alpha_q = 1$, T must lie exactly on the chord, approximating w^2 . If

$\alpha_q = 0$, then the inequalities become non-restrictive. We repeat the procedure described above for each of the squares

$$u_{i,j,1}^2, \quad u_{i,j,2}^2, \quad v_{l,k,1}^2, \quad v_{l,k,2}^2,$$

each time introducing a set of breakpoints $\{b_0, b_1, \dots, b_K\}$, binary variables α_q to select one interval, a continuous variable (say, $T_{u_{i,j,1}}, T_{u_{i,j,2}}, T_{v_{l,k,1}}, T_{v_{l,k,2}}$) to approximate the square, and linear constraints enforcing subinterval assignment and the linear chord approximation. Once these constraints and variables are added, the terms

$$u_{i,j,1}^2 \mapsto T_{u_{i,j,1}}, \quad u_{i,j,2}^2 \mapsto T_{u_{i,j,2}}, \quad v_{l,k,1}^2 \mapsto T_{v_{l,k,1}}, \quad v_{l,k,2}^2 \mapsto T_{v_{l,k,2}},$$

become linear references to the new continuous variables T within the system dynamics equations. Consequently, the expressions $u_{i,j,1}^2 - u_{i,j,2}^2$ and $v_{l,k,1}^2 - v_{l,k,2}^2$ can each be written in a linear form as:

$$(u_{i,j,1}^2 - u_{i,j,2}^2) \equiv (T_{u_{i,j,1}} - T_{u_{i,j,2}}), \quad (v_{l,k,1}^2 - v_{l,k,2}^2) \equiv (T_{v_{l,k,1}} - T_{v_{l,k,2}}).$$

By substituting these piecewise linear approximations into the previously derived CLEFO-based constraints, we obtain a fully linear Mixed-Integer Programming (MIP) model:

$$\left\{ \begin{array}{l} \min \sum_{i \in \mathcal{Y}} c_{1,i} Y_i + \sum_{j \in \mathcal{X}} c_{2,j} X_j \\ \left\{ \begin{array}{l} Y_i + \sum_{j \in \mathcal{Y}} \gamma_{i,j} Y_j + \sum_{j \in \mathcal{X}} \lambda_{i,j} (T_{u_{i,j,1}} - T_{u_{i,j,2}}) = \epsilon_i + \sum_{j \in \mathcal{X}} \beta_{i,j} X_j + \sum_{l \in \mathcal{X}} \sum_{k \in \mathcal{X}} \theta_{i,l,k} (T_{v_{l,k,1}} - T_{v_{l,k,2}}), \quad \forall i \in \mathcal{Y} \\ u_{i,j,1} = 0.5(X_j + Y_i), \quad u_{i,j,2} = 0.5(X_j - Y_i), \quad v_{l,k,1} = 0.5(X_l + X_k), \quad v_{l,k,2} = 0.5(X_l - X_k), \quad \forall i \in \mathcal{Y}, \quad \forall j, l, k \in \mathcal{X} \end{array} \right. \\ \text{s. t. } \left\{ \begin{array}{l} \sum_{q \in \mathcal{Q}} \alpha_{w,q} = 1, \quad b_{w,q-1} \alpha_{w,q} \leq w \leq b_{w,q} \alpha_{w,q}, \quad \alpha_{w,q} \in \{0,1\}, \quad \forall q \in \mathcal{Q}, \quad \forall w \in \{u_{i,j,1}, u_{i,j,2}, v_{l,k,1}, v_{l,k,2} | i \in \mathcal{Y} \text{ and } j, l, k \in \mathcal{X}\} \\ T_w - [b_{w,q-1}^2 + (b_{w,q} + b_{w,q-1})(w - b_{w,q-1})] \leq M(1 - \alpha_{w,q}), \quad \forall q \in \mathcal{Q}, \quad \forall w \in \{u_{i,j,1}, u_{i,j,2}, v_{l,k,1}, v_{l,k,2} | i \in \mathcal{Y} \text{ and } j, l, k \in \mathcal{X}\} \\ [b_{w,q-1}^2 + (b_{w,q} + b_{w,q-1})(w - b_{w,q-1})] - T_w \leq M(1 - \alpha_{w,q}), \quad \forall q \in \mathcal{Q}, \quad \forall w \in \{u_{i,j,1}, u_{i,j,2}, v_{l,k,1}, v_{l,k,2} | i \in \mathcal{Y} \text{ and } j, l, k \in \mathcal{X}\} \\ X_j^{\min} \leq X_j \leq X_j^{\max}, \quad Y_i^{\min} \leq Y_i \leq Y_i^{\max}, \quad \forall i \in \mathcal{Y}, \quad \forall j \in \mathcal{X} \end{array} \right. \end{array} \right.$$

The model above can be further simplified using a **Piecewise**($w, w_{\min}, w_{\max}, T_w, K$) operator that takes in the bounds $w_{\min}$ and $w_{\max}$ of the auxiliary variables $w \in \{u_{i,j,1}, u_{i,j,2}, v_{l,k,1}, v_{l,k,2} | i \in \mathcal{Y} \text{ and } j, l, k \in \mathcal{X}\}$ and the number of partitions K for the segmentation. Using these inputs, the operator then calculates the breakpoints $b_{w,q}$ and generates the constraint block for the piecewise linear approximation. The new model representation given these is as follows:

Equation 7-4 Single-unit mixed-integer linear programming optimization framework

$$\left\{ \begin{array}{l} \min \sum_{i \in \mathcal{Y}} c_{1,i} Y_i + \sum_{j \in \mathcal{X}} c_{2,j} X_j \\ s. t. \left\{ \begin{array}{l} Y_i + \sum_{j \in \mathcal{Y}} \gamma_{i,j} Y_j + \sum_{j \in \mathcal{X}} \lambda_{i,j} (T_{u_{i,j,1}} - T_{u_{i,j,2}}) = \epsilon_i + \sum_{j \in \mathcal{X}} \beta_{i,j} X_j + \sum_{l \in \mathcal{X}} \sum_{k \in \mathcal{X}} \theta_{i,l,k} (T_{v_{l,k,1}} - T_{v_{l,k,2}}), \quad \forall i \in \mathcal{Y} \\ u_{i,j,1} = 0.5(X_j + Y_i), \quad u_{i,j,2} = 0.5(X_j - Y_i), \quad v_{l,k,1} = 0.5(X_l + X_k), \quad v_{l,k,2} = 0.5(X_l - X_k), \quad \forall i \in \mathcal{Y}, \quad \forall j, l, k \in \mathcal{X} \\ \text{Piecewise}(w, w_{\min}, w_{\max}, T_w, K), \quad \forall w \in \{u_{i,j,1}, u_{i,j,2}, v_{l,k,1}, v_{l,k,2} | i \in \mathcal{Y} \text{ and } j, l, k \in \mathcal{X}\} \\ X_j^{\min} \leq X_j \leq X_j^{\max}, \quad Y_i^{\min} \leq Y_i \leq Y_i^{\max}, \quad \forall i \in \mathcal{Y}, \quad \forall j \in \mathcal{X} \end{array} \right. \end{array} \right.$$

7.5 Multi-Stage, Multi-Unit Network Optimization Framework

The single-unit CLEFO optimization model for wastewater treatment is extended to accommodate a multi-stage, multi-unit framework that captures the complexities of sequential treatment processes, which is required to solve the problem detailed in **Equation 3-2**. In this extended model, the wastewater undergoes S sequential treatment stages $s = 1, 2, \dots, S$, with each stage s capable of employing one among U_s alternative process units $u \in \mathcal{U}_s$. This selection is governed by binary decision variables $y_{s,u}$, ensuring that exactly one unit is chosen per stage, thereby constructing a superstructure of feasible unit configurations across all stages.

The model incorporates three distinct types of independent variables X namely, the influent compound concentrations (X^I), flow variables (X^F), and process-unit-specific variables (X^P). For the first stage ($s = 1$), influent compound concentrations $X_{s,i}^I$ represent the raw wastewater concentrations of compound $i \in \mathcal{I}$. For subsequent stages ($s > 1$), the influent concentrations are directly linked to the effluent concentrations from the preceding stage, i.e., $X_{s,i}^I = Y_{s-1,i}$. On the other hand, the value of volumetric flow variables X_f^F are decided exclusively at the first stage for $f \in \mathcal{F}$ are held constant across all stages to maintain consistent flow rates throughout the treatment process. Lastly, process-unit-specific variables $X_{u,j}^P$ and $j \in \mathcal{J}$ pertain to operational parameters unique to each process unit, such as aeration rates or chemical dosages – depending on the treatment mechanism of the unit. They are activated only if their corresponding unit u is selected in stage s .

The core of the optimization lies in the CLEFO-type balance equations for each unit u at each stage s and for each compound i . The balance equation is formulated as follows:

$$\begin{aligned}
Y_{s,u,i} &+ \sum_{i' \in \mathcal{I}} \gamma_{s,u,i,i'} Y_{s,u,i'} + \sum_{f \in \mathcal{F}} \lambda_{s,u,i,f}^{FY} (T_{s,f,i,1}^{FY} - T_{s,f,i,2}^{FY}) \\
&+ \sum_{j \in \mathcal{J}} \lambda_{s,u,i,j}^{PY} (T_{s,u,j,i,1}^{PY} - T_{s,u,j,i,2}^{PY}) \\
&+ \sum_{i' \in \mathcal{I}} \lambda_{s,u,i,i'}^{IY} (T_{s,i',i,1}^{IY} - T_{s,i',i,2}^{IY}) \\
&= y_{s,u} \epsilon_{s,u,i} + \sum_{f \in \mathcal{F}} \beta_{s,u,i,f}^F X_f^F + \sum_{j \in \mathcal{J}} \beta_{s,u,i,j}^P X_j^P + \sum_{i' \in \mathcal{I}} \beta_{s,u,i,i'}^I X_{i'}^I \\
&+ \sum_{f \in \mathcal{F}} \sum_{f' \in \mathcal{F}} \theta_{s,u,i,f,f'}^{FF} (T_{f,f',1}^{FF} - T_{f,f',2}^{FF}) \\
&+ \sum_{f \in \mathcal{F}} \sum_{j \in \mathcal{J}} \theta_{s,u,i,f,j}^{FP} (T_{u,f,j,1}^{FP} - T_{u,f,j,2}^{FP}) \\
&+ \sum_{f \in \mathcal{F}} \sum_{i' \in \mathcal{I}} \theta_{s,u,i,f,i'}^{FI} (T_{s,f,i',1}^{FI} - T_{s,f,i',2}^{FI}) \\
&+ \sum_{j \in \mathcal{J}} \sum_{j' \in \mathcal{J}} \theta_{s,u,i,j,j'}^{PP} (T_{u,j,j',1}^{PP} - T_{u,j,j',2}^{PP}) \\
&+ \sum_{j \in \mathcal{J}} \sum_{i' \in \mathcal{I}} \theta_{s,u,i,j,i'}^{PI} (T_{s,u,j,i',1}^{PI} - T_{s,u,j,i',2}^{PI}) \\
&+ \sum_{i' \in \mathcal{I}} \sum_{i'' \in \mathcal{I}} \theta_{s,u,i,i',i''}^{II} (T_{s,i',i'',1}^{II} - T_{s,i',i'',2}^{II}), \quad \forall s \in \mathcal{S}, u \in \mathcal{U}, i \in \mathcal{I}.
\end{aligned}$$

In this equation, $Y_{s,u,i}$ is the effluent concentration of compound i from unit u at stage s . The $\gamma_{s,u,i,i'} Y_{s,u,i'}$ term is the coupling between compounds i and i' . The λ terms represent the interaction between independent variables (flow f , process parameters j , influent concentrations i') and the effluent compound i , linearized through auxiliary variables T . The $y_{s,u} \epsilon_{s,u,i}$ term is the baseline intercept for compound i . When the unit u is not selected for stage s , the term $y_{s,u} \epsilon_{s,u,i} = 0$ so that the entire balance equation could be deactivated. The β terms are the direct effects of independent variables on compound i . The θ terms represent higher-order interactions among pairs of independent variables linearized via auxiliary variables T . The T variables are auxiliary variables representing the quadratic terms obtained from the linearization of the bilinear multiplications of the variables $Y_{s,i}$, X_f^F , X_j^P , and $X_{i'}^I$ illustrated as follows:

$$\begin{aligned}
X_f^F Y_{s,i} &\mapsto (u_{s,f,i,1}^{FY})^2 - (u_{s,f,i,2}^{FY})^2, \quad (u_{s,f,i,1}^{FY})^2 \mapsto T_{s,f,i,1}^{FY}, \quad (u_{s,f,i,2}^{FY})^2 \mapsto T_{s,f,i,2}^{FY}, \\
X_{u,j}^P Y_{s,i} &\mapsto (u_{s,u,j,i,1}^{PY})^2 - (u_{s,u,j,i,2}^{PY})^2, \quad (u_{s,u,j,i,1}^{PY})^2 \mapsto T_{s,u,j,i,1}^{PY}, \quad (u_{s,u,j,i,2}^{PY})^2 \mapsto T_{s,u,j,i,2}^{PY}, \\
X_{s,i'}^I Y_{s,i} &\mapsto (u_{s,i',i,1}^{IY})^2 - (u_{s,i',i,2}^{IY})^2, \quad (u_{s,i',i,1}^{IY})^2 \mapsto T_{s,i',i,1}^{IY}, \quad (u_{s,i',i,2}^{IY})^2 \mapsto T_{s,i',i,2}^{IY},
\end{aligned}$$

$$\begin{aligned}
X_f^F X_{f'}^F &\mapsto (v_{f,f',1}^{FF})^2 - (v_{f,f',2}^{FF})^2, & (v_{f,f',1}^{FF})^2 &\mapsto T_{f,f',1}^{FF}, & (v_{f,f',2}^{FF})^2 &\mapsto T_{f,f',2}^{FF}, \\
X_f^F X_{u,j}^P &\mapsto (v_{u,f,j,1}^{FP})^2 - (v_{u,f,j,2}^{FP})^2, & (v_{u,f,j,1}^{FP})^2 &\mapsto T_{u,f,j,1}^{FP}, & (v_{u,f,j,2}^{FP})^2 &\mapsto T_{u,f,j,2}^{FP}, \\
X_f^F X_{s,i'}^I &\mapsto (v_{s,f,i',1}^{FI})^2 - (v_{s,f,i',2}^{FI})^2, & (v_{s,f,i',1}^{FI})^2 &\mapsto T_{s,f,i',1}^{FI}, & (v_{s,f,i',2}^{FI})^2 &\mapsto T_{s,f,i',2}^{FI}, \\
X_{u,j}^P X_{u,j'}^P &\mapsto (v_{u,j,j',1}^{PP})^2 - (v_{u,j,j',2}^{PP})^2, & (v_{u,j,j',1}^{PP})^2 &\mapsto T_{u,j,j',1}^{PP}, & (v_{u,j,j',2}^{PP})^2 &\mapsto T_{u,j,j',2}^{PP}, \\
X_{u,j}^P X_{s,i'}^I &\mapsto (v_{s,u,j,i',1}^{PI})^2 - (v_{s,u,j,i',2}^{PI})^2, & (v_{s,u,j,i',1}^{PI})^2 &\mapsto T_{s,u,j,i',1}^{PI}, & (v_{s,u,j,i',2}^{PI})^2 &\mapsto T_{s,u,j,i',2}^{PI}, \\
X_{s,i'}^I X_{s,i''}^I &\mapsto (v_{s,i',i'',1}^{II})^2 - (v_{s,i',i'',2}^{II})^2, & (v_{s,i',i'',1}^{II})^2 &\mapsto T_{s,i',i'',1}^{II}, & (v_{s,i',i'',2}^{II})^2 &\mapsto T_{s,i',i'',2}^{II}.
\end{aligned}$$

The u and v auxiliary variables, whose square are represented by the T variables, are added using the same bilinear term relaxation protocol deployed in the single-unit method from the previous section. Their introduction adds the following constraints to the model:

$$\begin{aligned}
u_{s,f,i,1}^{FY} &= 0.5(X_f^F + Y_{s,i}), & u_{s,f,i,2}^{FY} &= 0.5(X_f^F - Y_{s,i}), \\
u_{s,u,j,i,1}^{PY} &= 0.5(X_{u,j}^P + Y_{s,i}), & u_{s,u,j,i,2}^{PY} &= 0.5(X_{u,j}^P - Y_{s,i}), \\
u_{s,i',i,1}^{IY} &= 0.5(X_{s,i'}^I + Y_{s,i}), & u_{s,i',i,2}^{IY} &= 0.5(X_{s,i'}^I - Y_{s,i}), \\
v_{f,f',1}^{FF} &= 0.5(X_f^F + X_{f'}^F), & v_{f,f',2}^{FF} &= 0.5(X_f^F - X_{f'}^F), \\
v_{u,f,j,1}^{FP} &= 0.5(X_f^F + X_{u,j}^P), & v_{u,f,j,2}^{FP} &= 0.5(X_f^F - X_{u,j}^P), \\
v_{s,f,i',1}^{FI} &= 0.5(X_f^F + X_{s,i'}^I), & v_{s,f,i',2}^{FI} &= 0.5(X_f^F - X_{s,i'}^I), \\
v_{u,j,j',1}^{PP} &= 0.5(X_{u,j}^P + X_{u,j'}^P), & v_{u,j,j',2}^{PP} &= 0.5(X_{u,j}^P - X_{u,j'}^P), \\
v_{s,u,j,i',1}^{PI} &= 0.5(X_{u,j}^P + X_{s,i'}^I), & v_{s,u,j,i',2}^{PI} &= 0.5(X_{u,j}^P - X_{s,i'}^I), \\
v_{s,i',i'',1}^{II} &= 0.5(X_{s,i'}^I + X_{s,i''}^I), & v_{s,i',i'',2}^{II} &= 0.5(X_{s,i'}^I - X_{s,i''}^I).
\end{aligned}$$

These variables have the following bounds:

$$\begin{aligned}
0.5(X_f^{F,min} + Y_{s,i}^{min}) &\leq u_{s,f,i,1}^{FY} \leq 0.5(X_f^{F,max} + Y_{s,i}^{max}), \\
0.5(X_f^{F,min} - Y_{s,i}^{max}) &\leq u_{s,f,i,2}^{FY} \leq 0.5(X_f^{F,max} - Y_{s,i}^{min}), \\
0.5(X_{u,j}^{P,min} + Y_{s,i}^{min}) &\leq u_{s,u,j,i,1}^{PY} \leq 0.5(X_{u,j}^{P,max} + Y_{s,i}^{max}), \\
0.5(X_{u,j}^{P,min} - Y_{s,i}^{max}) &\leq u_{s,u,j,i,2}^{PY} \leq 0.5(X_{u,j}^{P,max} - Y_{s,i}^{min}), \\
0.5(X_{s,i'}^{I,min} + Y_{s,i}^{min}) &\leq u_{s,i',i,1}^{IY} \leq 0.5(X_{s,i'}^{I,max} + Y_{s,i}^{max}),
\end{aligned}$$

$$\begin{aligned}
0.5 \left(X_{s,i'}^{I,min} - Y_{s,i}^{max} \right) &\leq u_{s,i',i,2}^{IY} \leq 0.5 \left(X_{s,i'}^{I,max} - Y_{s,i}^{min} \right), \\
0.5 \left(X_f^{F,min} + X_{f'}^{F,min} \right) &\leq v_{f,f',1}^{FF} \leq 0.5 \left(X_f^{F,max} + X_{f'}^{F,max} \right), \\
0.5 \left(X_f^{F,min} - X_{f'}^{F,max} \right) &\leq v_{f,f',2}^{FF} \leq 0.5 \left(X_f^{F,max} - X_{f'}^{F,min} \right), \\
0.5 \left(X_f^{F,min} + X_{u,j}^{P,min} \right) &\leq v_{u,f,j,1}^{FP} \leq 0.5 \left(X_f^{F,max} + X_{u,j}^{P,max} \right), \\
0.5 \left(X_f^{F,min} - X_{u,j}^{P,max} \right) &\leq v_{u,f,j,2}^{FP} \leq 0.5 \left(X_f^{F,max} - X_{u,j}^{P,min} \right), \\
0.5 \left(X_f^{F,min} + X_{s,i'}^{I,min} \right) &\leq v_{s,f,i',1}^{FI} \leq 0.5 \left(X_f^{F,max} + X_{s,i'}^{I,max} \right), \\
0.5 \left(X_f^{F,min} - X_{s,i'}^{I,max} \right) &\leq v_{s,f,i',2}^{FI} \leq 0.5 \left(X_f^{F,max} - X_{s,i'}^{I,min} \right), \\
0.5 \left(X_{u,j}^{P,min} + X_{u,j'}^{P,min} \right) &\leq v_{u,j,j',1}^{PP} \leq 0.5 \left(X_{u,j}^{P,max} + X_{u,j'}^{P,max} \right), \\
0.5 \left(X_{u,j}^{P,min} - X_{u,j'}^{P,max} \right) &\leq v_{u,j,j',2}^{PP} \leq 0.5 \left(X_{u,j}^{P,max} - X_{u,j'}^{P,min} \right), \\
0.5 \left(X_{u,j}^{P,min} + X_{s,i'}^{I,min} \right) &\leq v_{s,u,j,i',1}^{PI} \leq 0.5 \left(X_{u,j}^{P,max} + X_{s,i'}^{I,max} \right), \\
0.5 \left(X_{u,j}^{P,min} - X_{s,i'}^{I,max} \right) &\leq v_{s,u,j,i',2}^{PI} \leq 0.5 \left(X_{u,j}^{P,max} - X_{s,i'}^{I,min} \right), \\
0.5 \left(X_{i'}^{I,min} + X_{i''}^{I,min} \right) &\leq v_{s,i',i'',1}^{II} \leq 0.5 \left(X_{i'}^{I,max} + X_{i''}^{I,max} \right), \\
0.5 \left(X_{s,i'}^{I,min} - X_{s,i''}^{I,max} \right) &\leq v_{s,i',i'',2}^{II} \leq 0.5 \left(X_{s,i'}^{I,max} - X_{s,i''}^{I,min} \right),
\end{aligned}$$

We can generalize the bounding expressions above for the u and v variables as $w_{min} \leq w \leq w_{max}$. Using these bounds and a specified number of segments K , we can then execute the same piecewise linear approximation using the **Piecewise**($w, w_{min}, w_{max}, T_w, K$) operator as in the single-unit case for all auxiliary variables w . On the other hand, to ensure that the balance equations are active only when a specific unit u is selected in stage s , we incorporate Big- M constraints:

$$Y_{s,u,i} \leq M \cdot y_{s,u}, \quad X_{u,j}^P \leq X_{u,j}^{P,max} \cdot y_{s,u}, \quad \forall s \in \mathcal{S}, u \in \mathcal{U}_s, i \in \mathcal{I},$$

for a sufficiently large constant M . This effectively deactivates the equations for non-selected units, maintaining the integrity of the model. On the other hand, the objective function of this multi-stage model is designed to minimize the total costs associated with final effluent pollutant concentrations, operational costs of the selected units, and flow-related costs, potentially including penalties for raw influent concentrations. Formally, it can be expressed as:

$$\min \sum_{i \in \mathcal{I}} c_{1,i} \cdot Y_{S,i} + \sum_{s \in \mathcal{S}} \sum_{u \in \mathcal{U}_s} \sum_{f \in \mathcal{F}} c_{2,s,u,f}^F \cdot X_f^F \cdot y_{s,u} + \sum_{s \in \mathcal{S}} \sum_{u \in \mathcal{U}_s} \sum_{j \in \mathcal{J}} c_{2,s,u,j}^P \cdot X_{u,j}^P \cdot y_{s,u} + \sum_{i \in \mathcal{I}} c_{0,i} \cdot \Delta X_i^I.$$

Here, $\sum_{i \in \mathcal{I}} c_{1,i} \cdot Y_{S,i}$ penalizes the concentrations of pollutants in the final effluent. The term $\sum_{s,u,f} c_{2,s,u,f}^F \cdot X_f^F \cdot y_{s,u}$ accounts for operational costs related to flow variables, incurred only if the corresponding unit is selected. The term $\sum_{s,u,j} c_{2,s,u,j}^P \cdot X_{u,j}^P \cdot y_{s,u}$ incorporates costs associated with process-unit-specific operating parameters, active only upon unit selection. The term $\sum_{i \in \mathcal{I}} c_{0,i} \cdot \Delta X_i^I$ includes costs for raw wastewater amendments to alter compound concentration of the influent. The expressions $X_f^F \cdot y_{s,u}$ and $X_{u,j}^P \cdot y_{s,u}$ are nonlinear, but they can easily be linearized using McCormick envelopes, which is exact for the product of continuous and binary variables as follows:

$$\min \sum_{i \in \mathcal{I}} c_{1,i} \cdot Y_{S,i} + \sum_{s \in \mathcal{S}} \sum_{u \in \mathcal{U}_s} \sum_{f \in \mathcal{F}} c_{2,s,u,f}^F \cdot Z_{s,u,f}^F + \sum_{s \in \mathcal{S}} \sum_{u \in \mathcal{U}_s} \sum_{j \in \mathcal{J}} c_{2,s,u,j}^P \cdot Z_{s,u,j}^P + \sum_{i \in \mathcal{I}} c_{0,i} \cdot \Delta X_i^I,$$

$$X_f^F \cdot y_{s,u} \begin{cases} Z_{s,u,f}^F \leq X_f^{F,\max} \cdot y_{s,u} \\ Z_{s,u,f}^F \leq X_f^F \\ Z_{s,u,f}^F \geq X_f^F - X_f^{F,\max} \cdot (1 - y_{s,u}) \\ Z_{s,u,f}^F \geq 0 \end{cases} \quad \forall s \in \mathcal{S}, u \in \mathcal{U}_s, f \in \mathcal{F},$$

$$X_{u,j}^P \cdot y_{s,u} \begin{cases} Z_{s,u,j}^P \leq X_{u,j}^{P,\max} \cdot y_{s,u} \\ Z_{s,u,j}^P \leq X_{u,j}^P \\ Z_{s,u,j}^P \geq X_{u,j}^P - X_{u,j}^{P,\max} \cdot (1 - y_{s,u}) \\ Z_{s,u,j}^P \geq 0 \end{cases} \quad \forall s \in \mathcal{S}, u \in \mathcal{U}_s, j \in \mathcal{J}.$$

The other constraints on the objective are as follows: At each treatment stage $s \in \mathcal{S} = \{1, 2, \dots, S\}$, exactly one process unit $u \in \mathcal{U}_s = \{1, 2, \dots, U_s\}$ must be selected. This selection is governed by binary variables $y_{s,u} \in \{0, 1\}$:

$$\sum_{u \in \mathcal{U}_s} y_{s,u} = 1, \quad \forall s \in \mathcal{S}.$$

On the other hand, the effluent concentrations $Y_{s,i}$ at each stage s for each compound $i \in \mathcal{I} = \{1, 2, \dots, n\}$ are determined by the effluent of the selected unit within that stage. We enforce this rule through the following constraints:

$$Y_{s,i} \geq Y_{s,u,i} - M(1 - y_{s,u}), \quad \forall s \in \mathcal{S}, u \in \mathcal{U}_s, i \in \mathcal{I}.$$

$$Y_{s,i} \leq Y_{s,u,i} + M(1 - y_{s,u}), \quad \forall s \in \mathcal{S}, u \in \mathcal{U}_s, i \in \mathcal{I}.$$

The influent concentrations for each stage are directly linked to the effluent of the preceding stage. In the first stage, ΔX_i^I is the concentration added to the base concentration $X_i^{I,base}$ of the raw influent:

$$X_{1,i}^{I,min} \leq X_{1,i}^I \leq X_{1,i}^{I,max}, \quad X_{1,i}^{I,max} = X_i^{I,base} + \Delta X_i^I, \quad 0 \leq \Delta X_i^I \leq \Delta X_i^{I,max}, \quad \forall i \in \mathcal{I},$$

where ΔX_i^I is the decision variable while $X_i^{I,base}$ is given. For the generation of the breakpoints for the auxiliary variable corresponding to $X_{1,i}^I$, the maximum bound $X_i^{I,base} + \Delta X_i^{I,max}$ can be used as an approximate to not complicate the piecewise linearization scheme especially that ΔX_i^I is a decision variable. In the succeeding stages, we have:

$$X_{s,i}^I = Y_{s-1,i}, \quad \forall s = 2, \dots, S, \quad \forall i \in \mathcal{I}.$$

Flow variables X_f^F for $f \in \mathcal{F} = \{1, 2, \dots, F\}$ are decided only at the first stage and remain constant across all subsequent stages. Thus, X_f^F are decision variables $\forall f \in \mathcal{F}$. The same X_f^F from the first stage is used across all stages. All decision variables are bounded to reflect physical and operational limitations:

$$X_f^{F,min} \leq X_f^F \leq X_f^{F,max}, \quad \forall f \in \mathcal{F},$$

$$X_{u,j}^{P,min} \leq X_{u,j}^P \leq X_{u,j}^{P,max}, \quad \forall u \in \mathcal{U}_s, j \in \mathcal{J},$$

$$0 \leq Y_{s,u,i} \leq Y_{s,u,i}^{max}, \quad \forall s \in \mathcal{S}, u \in \mathcal{U}_s, i \in \mathcal{I}.$$

7.6 McCormick Envelopes

It is apparent that the following linearization employed on bilinear terms of the multi-stage, multi-unit framework makes the model large due to the multiple linearization steps of the method and its introduction of binary variables and constraints at each step, which begins by expressing the bilinear term xy as a difference of two squared auxiliary w variables, i.e., $(w_1)^2 - (w_2)^2$, and then the squared terms are linearized using piecewise linear approximation through the auxiliary T variables, i.e., $(w_1)^2 \mapsto T_1, (w_2)^2 \mapsto T_2$. For large applications, the linearization may render the model more computationally costly than its original nonlinear model. We address this issue by employing a linear relaxation using McCormick envelopes (McCormick, 1976). We know that x and y are continuous variables with known bounds

$$x^{min} \leq x \leq x^{max}, \quad y^{min} \leq y \leq y^{max}.$$

We introduce an auxiliary variable w to represent the product of the continuous variables $w = x \cdot y$. The standard McCormick linear constraints are:

$$\begin{cases} w \geq x^{\min}y + xy^{\min} - x^{\min}y^{\min} \\ w \geq x^{\max}y + xy^{\max} - x^{\max}y^{\max} \\ w \leq x^{\max}y + xy^{\min} - x^{\max}y^{\min} \\ w \leq x^{\min}y + xy^{\max} - x^{\min}y^{\max} \end{cases}$$

We perform this linear relaxation for all bilinear terms of the model: $X_f^F Y_{s,i}$, $X_{u,j}^P Y_{s,i}$, $X_{s,i'}^I Y_{s,i}$, $X_f^F X_{f'}^F$, $X_f^F X_{u,j}^P$, $X_f^F X_{s,i'}^I$, $X_{u,j}^P X_{u,j'}^P$, $X_{u,j}^P X_{s,i'}^I$, and $X_{s,i'}^I X_{s,i''}^I$. It should be noted that the McCormick method for the product of two continuous variables is not an exact linearization, but an approximation. The accuracy of the approximation is determined by the tightness of the bounds of the continuous variables. Given this, we suggest using the original linearized multi-stage, multi-unit model for small problems because it is near exact given a sufficiently large K for the piecewise linear approximation. For larger problems, however, the McCormick linearized multi-stage, multi-unit model is suggested for computational efficiency.

8 Methodology

8.1 Data

Bressani-Ribeiro et al. (2021)'s experimentally validated AQUASIM model is used to emulate a sponge-bed bio-tower. The model is used to simulate a thousand operational scenarios for influent volumetric flow rate, cross-sectional area, column height, sponge porosity, temperature, and influent compound concentrations. The bio-tower is divided into ten compartments. The compartment influent and compartment effluent compound concentrations are recorded for all compartments. At the end of the simulations, there are ten thousand data points that is used to calibrate the CLEFO model.

On the other hand, Fall et al. (2015)'s experimentally validated AQUASIM model is used to emulate a conventional activated sludge process. The model is used to simulate ten thousand operational scenarios for influent volumetric flow rate, recirculated activated sludge (RAS) flow, aeration tank volume, settler volume, and influent compound concentrations. At the end of the simulations, there are ten thousand data points that is used to calibrate the CLEFO model.

The Monte Carlo method is used to perturb the operational variables considered in both wastewater treatment technologies using a uniform distribution. The distribution bounds of the operational variables for each wastewater treatment technology are presented in **Table**.

On top of performing simulation-based case studies on the bio-tower and the conventional activated sludge model. We also perform a case study using actual data from 527 daily measures of sensors in an urban wastewater treatment plant employing a conventional activated sludge process from Poch (1993) of the University of California Irvine (UCI) Machine Learning Repository.

The main datasets are divided into calibration (70%) and validation (30%) sub-datasets. Mean-value imputation is executed for the dataset from Poch (1993). *StandardScaler* from Python's *sklearn* library is used to rescale the datasets. The rescaling is separately executed for each sub-dataset.

Lastly, cost data and other economics-related data are obtained from the literature. At present, the data is not yet curated, but such data should be manifold.

8.2 Comparison with Common Machine Learning (ML) Techniques

The CLEFO model is calibrated using ordinary least squares regression. The following ML models are calibrated using the same calibration datasets used for the CLEFO model:

- K-nearest Neighbors (KNN),
- Light Gradient-Boosting Machine (LightGBM),
- Artificial Neural Network (ANN),

- Random Forest (RF),
- Support Vector Machine (SVM),
- Extreme Gradient Boosting (XGBoost).

The calibrated models are then tested using the validation datasets, which they've never seen before. They are tasked to predict the effluent compound concentrations given the operational variables as the model inputs. The accuracy of their predictions is then compared. This comparison establishes whether CLEFO model is at par or better than common ML techniques at modeling wastewater treatment technologies.

Since CLEFO model is calibrated using ordinary least squares regression, we expect it to be the most computationally competitive technique compared to the common ML techniques, which are all notoriously computationally costly to implement, especially when the calibration dataset is large. Furthermore, we expect the CLEFO model to be more accurate at predicting effluent compound concentrations than the common ML techniques because it is modeled in view of the nature of microbial stoichiometry, which the black-box ML techniques do not consider.

8.3 Evaluating Optimization Models

The first evaluation is carried out between the linear and nonlinear single-unit CLEFO models. The second evaluation is carried out between the exact linear, McCormick, and nonlinear multi-stage multi-unit CLEFO models. The evaluations are carried out to compare the objective function values and computational gains from the linearization protocols at various model sizes. For the case of the single-unit CLEFO models, the model size is determined exclusively by the number of dependent and independent variables considered in the model. On the other hand, the model size of the multi-stage multi-unit CLEFO models is exclusively determined by the number of treatment stages, alternative wastewater treatment technologies per stage, dependent variable, and independent variables considered in the model.

We expect that the linear single-unit CLEFO model will be superior to its nonlinear counterpart on both objective function values and computational efficiency at all model sizes because there is not many variables or constraints involved when executing optimization on a single unit of wastewater treatment technology. Furthermore, although the linear single-unit CLEFO model has more decision variables (both continuous and binary) and constraints than its nonlinear counterpart, its linearity provides it with great computational advantages due to the existence of extremely powerful linear solvers.

On the other hand, we expect that the nonlinear multi-stage multi-unit CLEFO model will be superior to all its linear counterparts on computational efficiency when the model size is small, i.e., when there are only two stages and two process units per stage. The linear multi-stage multi-unit CLEFO model have a considerable number of binary variables, especially the exact linear model, which I expect will consistently be worse performing

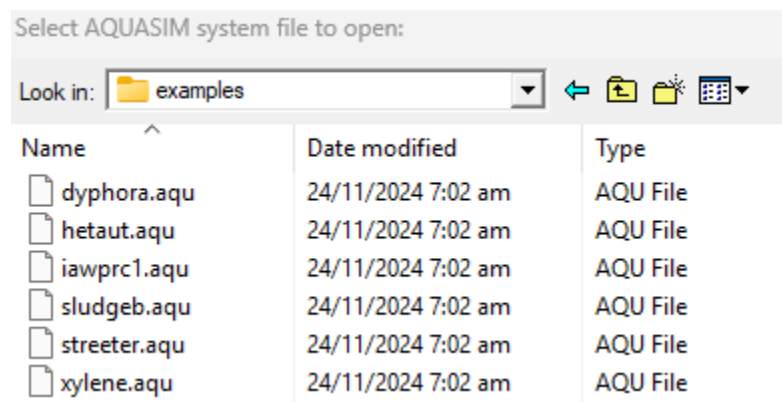
than either the nonlinear or the McCormick linear model because of the sheer number of its binary variables and constraints. However, we expect the McCormick linear multi-stage multi-unit CLEFO model to be superior on computational efficiency against the nonlinear and exact linear models when the model size is big. In most practical wastewater treatment design planning applications, there will be many wastewater treatment stages and alternative wastewater treatment process units per stage, which makes a superior model in such applications indispensable.

8.4 Application of Optimization Models

The single-unit CLEFO model can be easily applied to determine the optimal values for the operational design variables of sponge-bed bio-towers and activated sludge processes using the datasets described previously. First, the CLEFO model needs to be calibrated using the datasets, then the calibrated model is used for the single-unit optimization application.

The multi-stage, multi-unit CLEFO model requires that we have calibrated CLEFO models for all wastewater treatment process units considered in the design problem. To this end, we use the process units defined in the wastewater treatment case studies in the AQUASIM package presented in **Figure 8-1**. We assume that there are two treatment stages, i.e., primary treatment and secondary treatment stages. The first stage is either an activated sludge process which we define using datasets from Poch (1993) or an up-flow anaerobic sludge blanket which we define using datasets from *sludgeb.aqu*. The second stage is either a membrane bioreactor which we define using datasets from *xylene.aqu* or sponge-bed biotower which we define using datasets from Bressani-Ribeiro et al. (2021).

Figure 8-1 Wastewater treatment case studies in the AQUASIM package



The application of the single-unit or the multi-stage multi-unit CLEFO models are executed for different C:N:P ratios, i.e., 100:10:1, 500:5:1, and 50:10:1, and the best designs are obtained for each case.

8.5 Equipment and Software

For this research, an NVIDIA GeForce RTX 4090 graphics card is needed so that the manifold of computations carried out herewith are performed fast, especially during the model calibration and simulation phases detailed previously. Furthermore, all of the code in this study is developed using Python as programming language. The *pyomo* library is used for implementing all optimization models and are all saved as Mathematical Programming System (mps) files. On the other hand, the *sklearn* library is used for the implementation of the common ML techniques.

The High-performance parallel linear optimization software (HiGHS) algorithm is used to solve all linear optimization models, while the Interior Point Optimizer (Ipopt) algorithm is used to solve all nonlinear model models.

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10 Budgetary Requirements

	Cost	Needed Qty.	Price/SKU	Stock Keeping Unit (SKU)	Total Qty.	Unit
Summary						
Covered by ERDT scholarship	350,000.00					
Total cost	350,000.00					
Delivery fee (assumed to be 10% of materials cost)	24,422.6					
Subtotal: Computing Equipments	248,977.4					
Subtotal: other costs	76,600.00					
Other costs						
Comprehensive exam fee	5,000.00	1.00	5,000.00	1.00	1.00	ea
Proposal hearing fee	25,000.00	1.00	25,000.00	1.00	1.00	ea
Final defense fee	25,000.00	1.00	25,000.00	1.00	1.00	ea
Ring bound copies printing fee (during proposal hearing and final defense)	3,600.00	12.00	300.00	1.00	12.00	ea
Book printing fee	18,000.00	30.00	600.00	1.00	30.00	ea
Computing Equipments						
Intel® Core™ i9-14900KF, 24 Cores & 32 Threads Unlocked Gaming Processor	24,258.83	1	24,258.83	1.00	1.00	ea
MSI PRO Z790-P WiFi ProSeries Motherboard (USB 3.2 Gen2, Wi-Fi 6E)	11,105.87	1	11,105.87	1.00	1.00	ea
Cooler Master i70C ARGB Intel Low-Profile 37.08 CFM CPU Air Cooler	1,373.11	1	1,373.11	1.00	1.00	ea
NZXT H6 Flow RGB CC-H61FB-R1 Compact Dual-Chamber Mid-Tower Airflow Case Includes 3 x 120mm RGB Fans Panoramic Glass Panels High-Performance Airflow Panels Cable Management Black	7,822.45	1	7,822.45	1.00	1.00	ea

MSI Gaming GeForce RTX 4090 24GB GDDR6X 384-Bit HDMI/DP Nvlink Tri-Frozr 3 Ada Lovelace Architecture OC Graphics Card	169,402.59	1	169,402.59	1.00	1.00	ea
G.SKILL Trident Z5 RGB Series (Intel XMP 3.0) DDR5 RAM 48GB (2x24GB) 8200MT/s CL40-52-52-131 1.35V Desktop Computer Memory UDIMM - Matte White (F5-8200J4052F24GX2-TZ5RW)	14,613.17	1	14,613.17	1.00	1.00	ea
Samsung EVO 970 1TB PCIe Gen3 x4 NVMe M.2-2280 Internal Solid State Drive with V-NAND Technology & 1024MB Cache	9,294.35	1	9,294.35	1.00	1.00	ea
EVGA SuperNOVA 1000 G6, 80 Plus Gold 1000W, Fully Modular, Eco Mode with FDB Fan, 10 Year Warranty, Includes Power ON Self Tester, Compact 140mm Size, Power Supply 220-G6-1000-X1	11,107.03	1	11,107.03	1.00	1.00	ea

11 Timeline of Activities with Expected Outputs

2025	January	February	March	April	May	June	July
Coding all models in Python							
Performing all simulations and gathering all data							
Writing journal articles							
Finalizing dissertation							
Defense and revisions							